



MONTRAN

National Bank of Georgia

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Inception Report

Instant Payments System

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Introduction

National Bank of Georgia (IPS Operator) has contracted Montran for the implementation of an Instant Payments solution, based on the Montran Instant Payments System (IPS), in accordance with the accepted technical characteristics, part of the agreed Contract between the parties. The name of the solution is determined by the IPS Operator, and the IPS acronym is used to refer to the solution described in this document.

For the purpose of the signed contract, the IPS Operator is the same as the Proxy operator. However, according to the architecture agreed upon, the IPS and the Proxy will be accessible via two separate URLs. That means one Module for Maintenance and Control (the Graphic User Interface) for the IPS and one Module for Maintenance and Control (the Graphic User Interface) for the Proxy. Participants will be registered twice, once in IPS and second in Proxy, or just in Proxy.

Specifications described in this document and its related annexes are common for both IPS and Proxy. Given the two URL accesses, some information will be duplicated in the MMCs. This document addresses Proxy only related specific details in paragraphs dedicated to the Proxy.

Definitions and Acronyms

TERM	ACR	DESCRIPTION
Acronym	ACR	Acronyms used in this document.
Available Balance	AB	Value automatically calculated by the IPS, Net Balance COMPLETE (Credit - Debit) – HOLD Debit Balance . Each IPS participant can see its available balance. The IPS Operator can see the available balance of all IPS Participants. The available balance takes into consideration pending debits, meaning that a pending debit (status HOLD) is subtracted from the displayed available balance. That pending debit is a pending credit for the beneficiary participant. The AB displayed for the beneficiary participant will not calculate the pending credit (status HOLD). The AB displayed for the beneficiary participant will contain (have the added credit value) the credit only after the transaction has status COMPLETE.
Business Identifier Code	BIC	BIC is the International ISO standard ISO 9362. This standard specifies the elements and structure of a universal identifier code, the business identifier code (BIC), for financial and non-financial institutions, for which such an international identifier is required to facilitate automated processing of information.

TERM	ACR	DESCRIPTION
Certificate Authority	CA	An authorized authority which issues the certificates used by the IPS and registered by participants for SSL communication and Digital Signature validation.
Certificate Revocation List	CRL	To verify the revocation of Participants' digital certificates (user certificates and STP applications), Montran solutions use Certificate Revocation Lists published by the Certificate Authority that issue those certificates.
Currency	CCY	Currency code
Daily Rollover		Event from Timetable marking the change of business day in the system. Typically scheduled at 00:00 but can be done manually in case of contingency.
Discrete Event		A discrete event is a single liquidity adjustment or daily rollover event scheduled automatically or manually. This event occurs only in IPS.
Event Window		An event window is a continuous interval of time in which the process or scheduled event is executed repeatedly with a configurable cycle time. This event window occurs only in IPS.
Four-eyes Check	FeC	Mechanism to prevent unauthorized operations in the IPS. Any addition, modification, removal, blocking, and unblocking will take effect in the IPS only when a second operator approves the operation.
HOLD		Status of transaction for which funds are not yet available. Until the transaction is completed or cancelled the amount is not available either for the debtor or the creditor.
Instant Payments System	IPS	The Instant Payments system developed by Montran and described in this document.
Internet Protocol	IP	The method or protocol by which data is sent from one computer to another on the internet.
IPS Direct Participant	DP	IPS Direct Participant (settlement: direct, clearing: direct), which is directly connected to IPS through dedicated VPN and has an account in RTGS system. IPS Direct Participant manages liquidity separately for each IPS indirect Participant as well as for IPS Non- Settlement Direct Clearing Participant to which it provides a settlement service.
IPS Indirect Participant	IP	IPS Indirect Participant (settlement: indirect, clearing: indirect) which has no direct connection to IPS. This participant uses other IPS Direct Participants for clearing and settlement of transactions.
IPS Non-clearing Direct Settlement Participant	NCDSP	IPS Non-clearing Direct Settlement Participant (settlement: direct, clearing: indirect) which is not directly connected to IPS, but settlement is done through its own account in the RTGS system.
IPS Non-settlement	NSDCP	Non-settlement Direct Clearing Participant (settlement: indirect, clearing: direct) which is directly connected to IPS

TERM	ACR	DESCRIPTION
Direct Clearing Participant		but settlement is done through another Direct Settlement Participant.
IPS Participant		Institution participating in the Instant Payments System.
Proxy Participant		Participant registered only in the proxy module
Proxy IPS Participant		Participant registered both in the proxy module and the IPS.
Proxy Direct Participant	DP	Proxy Direct Participant (settlement: direct, clearing: direct), which is directly connected to Proxy through dedicated VPN and has an account in RTGS system.
Proxy Non-settlement Direct Clearing Participant	NSDCP	Proxy Non-settlement Direct Clearing Participant (settlement: indirect, clearing: direct) which is directly connected to Proxy, but settlement is done through another Proxy Direct Settlement Participant.
Message		The messages received, processed and sent by the IPS are: - Payment Confirmation/Rejection (pacs.002) - Recall request (camt.056) and positive (pacs.004)/negative (camt.029) answer to recall request - Request to pay (pain.013) and positive/negative answer (pain.014) to RTP - Request for cancelation (RfC) and request for status update of RfC (camt.055) - Investigation (pacs.028) - Payment initiation (pain.001) and status for initiation (pain.002) - Reconciliation (camt.053) - Balance snapshot (positions.001) - Participant List Table (participants.001) - Credit Notification (between RTGS and IPS, presented in Annex F – RTGS Interface) (camt.054) - Participant Instant Credit Transfer (within IPS, presented in Annex D – Participant Interface) (pacs.008) - Participant Credit Transfer (within IPS, presented in Annex D – Participant Interface) (pacs.009) - Participant Credit Transfer (between RTGS and IPS, presented in Annex F – RTGS Interface) (pacs.009) - IPS Event Notification (between RTGS and IPS, presented in Annex F – RTGS Interface) (admi.004) - Balance report (between RTGS and IPS, presented in Annex F – RTGS Interface) (camt.052) The messages received, processed and sent by the Proxy are: - Identification Modification Advice (acmt.022) - Identification Verification Request (acmt.023) - Identification Verification Report (acmt.024) - Payment Status Report (pacs.002) - Participant Credit Transfer (between RTGS and Proxy, presented in Annex F – RTGS Interface)

TERM	ACR	DESCRIPTION
		Proxy Event Notification (acmt.022)
Message Processing Duration	MPD	The period of time from the moment when the message is received in the IPS and the moment when the IPS delivers a message to the next Participant in the processing chain. The concept of Message Processing Duration is used for the performance report where it is irrelevant to calculate the message process time outside of IPS. The total time for the processing of a transaction, including the time spent outside of IPS, can be seen in the menu Transactions -> List, fields: Origination Time, Process Completion Time.
Module for Management and Control	MMC	The component of Montran's IPS solution that implements the user interface and allows system management operations.
Offline Participant Period	OPP	The time period during which the system Participant is not connected to the system and cannot receive payment messages.
Operator		Entity that operates and manages the IPS including but not limited to monitoring and control of the IPS operation. The Operator role belongs to the Purchaser unless the Purchaser decides to transfer the Operator role to another designated entity. The transfer of such Operator role includes the same conditions as the Purchaser owns it. (NBG for now) For clear differentiation, the documents will use the following terms: -Operator as the institution that operates and manages the IPS. -Operator User as the person employed by the Operator, who manages functionalities of the IPS, depending on the granted access rights (defined through profile and user groups).
Participant		-Participant as the institution registered in the IPS that sends/receives payment messages. The Participant is of one of the four types defined in this table: IPS Direct Participant, IPS Indirect Participant, IPS Non-settlement Direct Clearing Participant, and IPS Non-Clearing Direct Settlement Participant. -Participant User as the person employed by that Participant, who manages functionalities of the IPS, depending on the granted access rights (defined through profile and user groups). The definition of Participant and Participant user, as institution and person employed by that institution, applies to the proxy module as well.
Processing Module	PM	The component of Montran's IPS solution that processes the messages received from Participants.

TERM	ACR	DESCRIPTION
Real-Time Gross Settlement	RTGS	Real Time Gross Settlement system, operated and managed by the RTGS Operator.
Request-To-Pay	RTP	Based SEPA Request to Pay (SRTP) scheme rulebook of the European Payments Council (EPC) that can be defined as part of the business model of the IPS application for the IPS Operator.
SEPA Instant Credit Transfer Schema	SCT Inst	The SEPA Instant Credit Transfer Schema of the European Payments Council.
Settlement Window		The Settlement Window represents the time period in which IPS can communicate with the RTGS. Only within this time frame can funds be moved from and to the RTGS pool account, which is mirrored in the IPS as MPFA. The settlement window is defined according to the values set in Timetable menu: Settlement Window Start and Settlement Window Stop fields which display the time. Those values can be either set automatically through notifications from RTGS or manually by the IPS Operator. In case of contingency, the actual execution of the settlement window start and stop events can be done manually.
Settlement window start		Event from Timetable marking the beginning of the settlement window (can be scheduled or manual)
Settlement window stop		Event from Timetable marking the end of the settlement window (can be scheduled or manual)
Straight-Through-Processing	STP	Straight-Through-Processing interface that allows automatic access.
Sweep Event		The moment when IPS initiates the needed liquidity adjustments.
System		System is the Instant Payments System described by the Inception Report. IPS or system refer to the same described solution.
Technical Accounts	TA	The account of an IPS Participant (any type of Participant), created and managed in IPS, calculates and manages the Participant's balances. We will use the term sub-account to refer to the accounts of the indirect settlement participants. Functionally there is only one type of account in IPS (=IPS technical account)
Third Party Service Provider	TPSP	Third Party Service provider used for providing connectivity services to IPS Indirect Clearing participants.
Transaction Type Code	TTC	The Transaction Type Code (TTC) used in the billing settlement messages.
IPS User		A person employed by the IPS Operator or any IPS Participant, accessing some of the IPS's functionality, depending on the access rights granted.

TERM	ACR	DESCRIPTION
Proxy User		A person employed by the Proxy Operator or any Proxy Participant, accessing Proxy functionality, depending on the access rights granted.
IPS Condition		Internal structure used by the application to differentiate between HOLD and COMPLETE amounts for each IPS account.

Further References Documents

NAME	AUTHOR
Supply, Installation, Integration, Testing and Commissioning of an Instant Payment System (IPS) and Related Services	IPS Operator
SEPA Instant Credit Transfer Scheme Rulebook	EPC
SEPA Instant Credit Transfer Scheme Inter-PSP Implementation Guidelines	EPC
SEPA Request-To-Pay (SRTP) Scheme Rulebook	EPC
SEPA Request-To-Pay (SRTP) Inter-RTP Service Provider SRTP Implementation Guidelines	EPC

General Description of the IPS Solution

The Montran IPS solution ensures the provision of the Instant Payment service according to the SCTInst and SEPA RTP payment schema, adapted for the local currency. It allows IPS participants, including PSP and commercial banks, to provide their customers with secure instant payment 24/7/365 for which the transfer is irrevocable and final. It also allows authorized entities as proxy participants to have access to the proxy services.

Montran IPS General Architecture

The Montran IPS solution is made of two major modules:

1. **The Processing Module (PM)** is the application that processes the messages from Participants. It runs on multiple nodes (real or virtual servers) to ensure operational continuity. The Processing Module does not provide a user graphical interface. The PM provides only two STP interfaces used by other applications such as the Module for Management and Control (MMC) and the Straight-Through-Processing (STP) applications from Participants:
 - a. **The STP Interface for Participants** provides features for sending and receiving messages, queries for net position and clearing limit, etc.
 - b. **The STP Interface for administration** is used only by the Module for Management and Control for implementing management tasks: updating Participants, accounts, settlement processing, monitoring, etc.
2. **The Module for Management and Control (MMC)** enables the IPS Operator's users to manage the entire IPS solution. It provides a Web-based graphical interface for accessing all features for the Operator's users and users from Participants. The management, monitoring and reporting functions provided by the management module (MMC) are described in the section Functional Specifications of the IPS Solution.

The Proxy Architecture is described in Annex E – Proxy Interface.

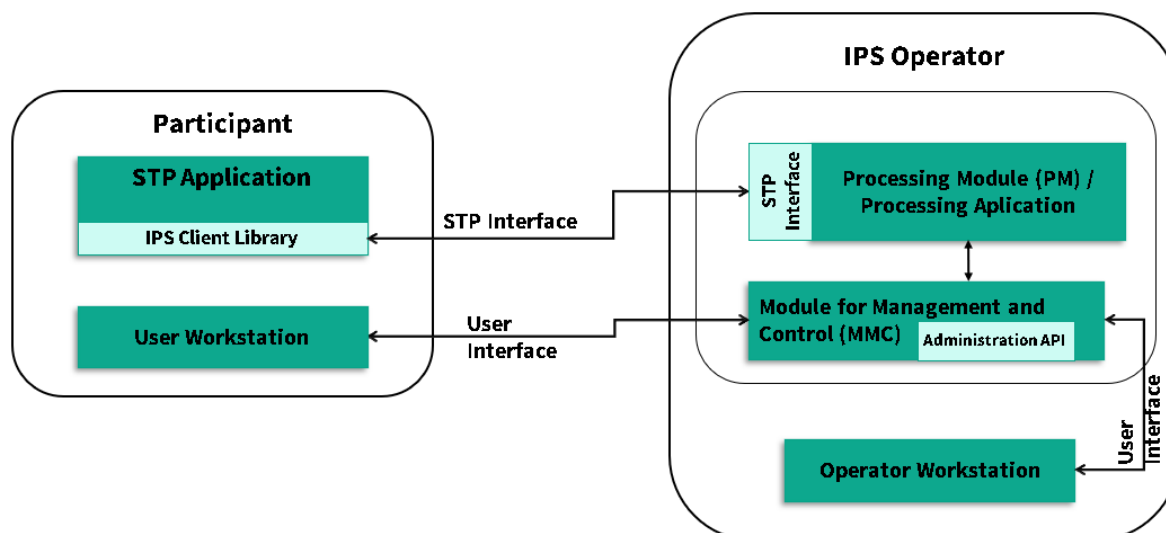


FIGURE 1. IPS ARCHITECTURE

Access to IPS solution is provided for Participants through:

- **IPS messaging interface:** an API exposed by the processing nodes through which the participants can initiate payments, recalls, request to pay, proxy management and queries, receive notifications for these flows, receive reconciliation messages
- **The MMC user interface:** the MMC serves both as a Participant Portal and as a Management Portal for the IPS and the Proxy depending on the user group and profile of the authenticated user.

The MMC was designed following the best practices in UI and UX, by a dedicated UX team. An accessibility audit was also performed to ensure usability based on colors and font sizes. It is continuously updated, and any usage errors or newly defined best practices can be implemented. The MMC supports the industry leading browsers, including Microsoft Edge, Google Chrome and Firefox. MMC event timer of liquidity adjustments, listing operations, reporting, and the MMC restart will have no impact on the PM component (which executes transactions) availability and performance.

Montran will install, configure and provide support for three testing environments. Installation and support from Montran for additional testing environments can be done as a change request.

Interfaces for Automatic Access (STP)

The Straight-Through-Processing (STP) interfaces of the IPS solution are divided into two categories:

1. **STP interface for Participants** – This interface is used by the Participants' applications to send and receive messages.
2. **Internal interfaces** – This category includes the interfaces required for communication with the following systems:

- a. RTGS for liquidity adjustment events or scheduled operations and for billing for the IPS, and RTGS for billing only for the Proxy. The interface details and the messages used are described in the IPS-RTGS Interface Annex.
- b. Email server for sending reports and alerts by email using the SMTP protocol.
- c. SMS gateway for sending alerts by text message.

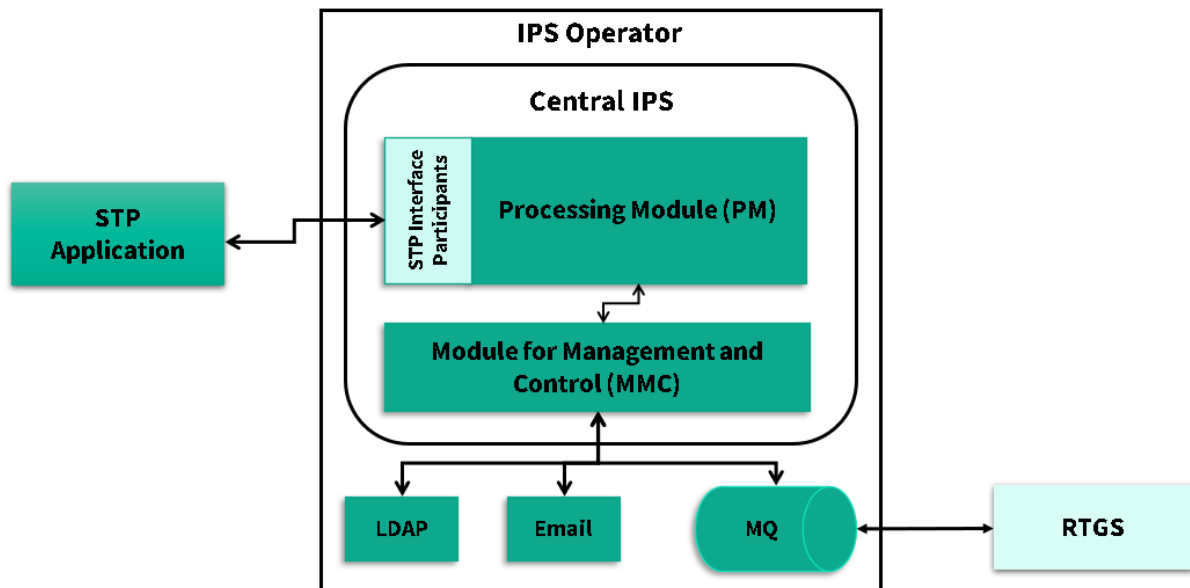


FIGURE 2. STP INTERFACE

Functional Specifications of the IPS Solution

Functional Features

The following features are part of the IPS solution:

Currency settings	The IPS is configured to use the following currency GEL.
Reports	IPS users can filter the reports generated by IPS during specific events (saved reports) or the on-demand reports. The IPS Operator users can see the consolidated reports that contain the information for all Participants. Direct Participant's users can view only the reports that belong to their institution in case of participants that do not provide services as direct participant for other types of participants (IP, NCDSP, NSDCP). Direct participants that offer DP services to other participants can see - Data belonging to IP, NSDCP for which they act as a direct settlement participant in Billing and Liquidity Adjustment reports. - Data belonging to IP, NCDSP for which they act as a direct clearing participant in Reconciliation, Processing Exception reports The participants of type IP, NCDSP, or NSDCP can only see the reports that belong to their institution. Direct settlement participants will not be able to see indirect settlement participants messages/transactions. Direct clearing participants will see their indirect clearing participants messages/transactions. Indirect clearing participants will only see their own messages/transactions. If the indirect clearing participant moves to another direct clearing participant, the indirect clearing participant will see their own old messages/transactions, but the new direct participant will only see the messages/transactions made after the switch. The reports can be exported in the following formats: PDF, XLS, XLSX, XML, CSV. Older generated reports can be filtered and displayed from the MMC. The solution is capable of displaying data from archive database as well as from main database for the retention period, for saved reports. If the report entries are archived then they will be available in the interface. On demand (= ad hoc) reports are generated at the moment the user clicks on the corresponding action in MMC. Depending on the report type there are filters for business day or time interval for which data to be included in the report.

	On demand (ad hoc) reports will not affect the transaction processing performance which is done on PM component but might affect MMC performance (which is used for RTGS communication, DMS module, IPS system maintenance, monitoring etc.). Regarding access rights for on-demand reports, they can also be controlled through profile functionality. Custom profiles can be created with on-demand reports removed (completely or partially) from the functionality list. The specifications of the reports are provided in Annex A - Reports.
List transactions	Users can filter the list of transactions that were processed by IPS. IPS Operator users can filter all transactions in the system. Participants' users can filter only transactions where they are the sender or the receiver party. The IPS displays detailed information about each processed transaction, including: sender, receiver, debtor, creditor, origination time, receive time, forward time, process completion time, status, error code. The list of transactions can be exported into PDF and CSV format.
User Sessions	IPS Operator users can filter the users who are currently logged into the system. IPS Operator users can get a list with the logged users in the system.
Limits	IPS Participants can set their limits in the IPS. The limits act as the reference levels for the account position sweeping for the purpose of liquidity management. After the minimum level, is breached, the sweeping will trigger an adjustment up to base balance – via RTGS credit (pacs.009). After the maximum level is breached, the sweeping will trigger an adjustment down to base balance– via RTGS credit (pacs.009). A Before Holiday Limits increment is also present, allowing to provide extra liquidity for the holidays by adding it to the limits. The Before Holiday Limits increment will be activated for the window event of the day preceding the bank holiday. List: Participants' users can list their limits. Set: Participants' users can set the amount for the limit. Approve: Participants' users can approve the changes performed over the limits.
User Activity	User activity is stored in the system database. The historical records can be filtered with this function. The stored information about user activity includes Username, Participant, Workflow (menu entry), Function (specific operation such as ReconciliationReportGenerate, ListReveal, GetPositions etc.), Operation (operation type –

	List, Enter, Modify, Disable, Remove, Validate, Approve, Reject, Close, Claims, Reopen, Reassign, Block, Unblock, Activate, Execute, Cancel, View), IP Address, Start time, Execution time. List: IPS Operator users can list the activity details of all users in the system. Participants' users can list the activity details of all users who belong to their institution. In case of filter for messages and transactions, the system also stores and displays the filter values selected by the user.
Messages	Users can filter the list of messages that were sent/received by IPS and export them into PDF and CSV format IPS Operator users can filter all messages in the IPS . Participants' users can filter only messages that they sent or received. List: IPS Operator users can list all messages in the IPS . Participants' users can list messages that were sent or received by their institution. The IPS displays detailed information about each sent or received message, including: sender, receiver, reference, status, processed time, destination queue (if applicable). The schemas employed for the ISO20022 support UTF-8 by default, allowing all text in English and Georgian to be exchanged. The symbol „Ⴀ“ is compatible with the UTF-008 encoding. All messages processed by IPS, including the Proxy module, are stored in the main database and can be queried in MMC.
Audits	IPS keeps track of all modifications for the entities managed in MMC. This history can be seen for each entity in the reveal (modify) menu screen, after clicking on “Audit Details” button. IPS display a list of all modifications (operation – create/modify/approve/etc., username, timestamp). The user can click on a row to see full details of the modified entity (the values before and after the operation).
External Codes	IPS allows the Operator to list and manage the External Codes used for validation of API messages from participants. The operator can create or remove individual codes through the user interface (Management – External Codes menu in MMC) or upload a complete list of codes (which will replace the existing codes). The code types managed through this functionality are: - Category Purpose (PmtTpInf – CtgyPurp – Cd) - Local Instrument (PmtTpInf – LclInstrm – Cd) - Clearing System (SttlmInf – ClrSys - Cd) - Service Level (SttlmInf – SvcLvl - Cd) - Purpose (Purp – Cd)

	Recall Reasons (camt.056, CxlRsn – Rsn – Cd) Recall Nack Reasons (camt.029, CxlStsRsnInf – Rsn – Cd) Return Reasons (pacs.004, RtrRsnInf – Rsn – Cd) Reject Reasons (pacs.002, StsRsnInf – Rsn – Cd) Reason Codes for Request for cancellation (camt.055, Rsn, Undrlyg - OrgnPmtInfAndCxl - TxInf - CxlRsnInf)
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3.1.1. Participants

System Participants are payment institutions that have access to IPS. The introduction and modification of the data associated with a Participant in the IPS is made by the IPS Operator users.

Participant Identification Details (IPS)

The IPS allows the IPS Operator to record the following information for each Participant.

NAME/ DESCRIPTION	FORMAT
Participant name (official name)	Text, max 100
Abbreviated name	Text, max 50
BIC code	Text, 8 or 11, BIC validation The BIC is validated with the following criteria: - alphanumeric characters, uppercase letters - size 8 or 11 - in case of size 11 then it must end with 'XXX' string - the BIC is not already taken by another registered participant
Preferred language with English options for generating automated reports	Combo Box control type for selection
Contact details: Name and telephone number	Text, max 60 each, optional
E-mail address for receiving automatically generated reports	Text, max 100, e-mail address formatting validation, optional
Validity Business date interval during which participant is set active in the system. If « from » is set then the participant will be created as DISABLED and will be activated automatically at the specified date.	Date selector, optional

If « to » is set then the participant will be automatically set to DISABLED at the beginning of the specified date.	
Participant's address	Text, max 500, optional
Third Party Service Provider Used to identify non-clearing participants, with the following options: DIRECT, Indirect clearing by selection of the direct participant	Combo box control type (DIRECT by default)
Settlement type (whether the participant has an account in RTGS or is an indirect-settlement participant) with the following options: Direct settlement Indirect settlement by selection of the direct depository associated partner	Combo Box control type (DIRECT by default)
Pooled liquidity for IP This flag can be set for indirect settlement participants. Checking this flag will have the effect that the direct participant will share its IPS liquidity (a single account in IPS). If the flag is unchecked, then liquidity sharing is disabled. Indirect settlement participants will have separate IPS accounts from their direct participant account with their own low/base/high limits (funds are transferred from the direct participant's RTGS account).	Checkbox, optional
Schema Payment Mapping – The selected payment schemas which will be used to evaluate transaction limits, timeouts.	Checkbox
Registered in Proxy Flag to indicate if this participant is also registered in the Proxy Module In IPS, it is set only for information purposes, and the actual registration must be done in the Proxy Module interface.	- Checkbox -
Not Billed	Checkbox
Reconciliation message Configuration per participant whether to receive reconciliation messages (camt.053) and through which mechanism	Combo box: - None - Cycle - End of Day Both
Payment Initiation Service Provider (PISP) Whether this Participant is allowed to send payment initiation messages.	Checkbox
PISP Participants If this participant is marked as a PISP, then this is the list of participants on behalf of whom it can initiate payments.	Multi select Combo Box

Participant Identification Details (Proxy)

NAME/ DESCRIPTION	FORMAT
Participant name (official name)	Text, max 100
Abbreviated name	Text, max 50
BIC code	Text, 8 or 11, BIC validation The BIC is validated with the following criteria: - alphanumeric characters, uppercase letters - size 8 or 11 - in case of size 11 then it must end with 'XXX' string - the BIC is not already taken by another registered participant
Preferred language with English options for generating automated reports	Combo Box control type for selection
Contact details: Name and telephone number	Text, max 60 each, optional
E-mail address for receiving automatically generated reports	Text, max 100, e-mail address formatting validation, optional
Validity Business date interval during which participant is set active in the system. If « from » is set then the participant will be created as DISABLED and will be activated automatically at the specified date. If « to » is set then the participant will be automatically set to DISABLED at the beginning of the specified date.	Date selector, optional
Participant's address	Text, max 500, optional
Settlement type (whether the participant has an account in RTGS or is an indirect-settlement participant with the following options: Direct settlement Indirect settlement by selection of the direct depository associated partner	Combo Box control type (DIRECT by default)
Proxy Service Participation The proxy service functionality allowed for this participant.	Multi select Combo Box - Standard (proxy management operations) - Proxy Lookup - Account Holder Inquiry - Reachability Check - Account Possession Check - Verification of Payee
Proxy Participant Type	Combo Box - Bank Institution

NAME/ DESCRIPTION	FORMAT
	- Non-bank Institution - Authorized Entity - Account Data Provider
Proxy Service Enabled Aliases A list of aliases enabled for the participant.	Multi select Combo Box (mobile number, email address, merchant id, identification number)
Not Billed	Checkbox
Send notification for manual operations Whether the system sends notification messages on API after manual operations in the user interface	Checkbox

The completion of this information is mandatory, except for:

- Contact data
- Address
- Email address.

The IPS must prevent the existence of Participant records containing the same values (duplicate values) for each of the following fields, considered as identification fields:

- Participant name (official name)
- Abbreviated name
- The BIC code
- Associated group name

Operations with Participants

In addition to the default **List**, **Create**, **Modify**, **Remove**, and **Approve** operations applicable to all entities managed by the system, Participants also support the following operations:

- List:** Users in the IPS can list the existing Participants in the system.
- Create:** the IPS Operator users can create a new Participant in IPS. This operation requires an approval step. After the creation and approval of a new Participant, IPS will create automatically the technical accounts of the Participant. The time of the Participant becoming available can be controlled by the operator, in no less than the next business day.
- Approve:** the IPS Operator users can approve the changes performed in the following operations: create, modify, block, unblock, disable, remove. Once the approval step is performed, the changes are permanent (meaning that the operation is registered in entry history and it's persisted in the database even if the application crashes/restarts etc.) and are applied immediately. Additional operations on that entity are still permitted after approve.
- Modify:** the IPS Operator users can modify the details of an existing Participant, including the BIC. This operation requires an approval step. All mappings in the application are done

through database id's, i.e. accounts, transactions, digital certificates are mapped to an id in the participant table. Therefore, even if the BIC changes the mapping stays the same (the participant will still be able to see its old data) but future validations will take into account the new BIC (the participant would not be able to initiate transactions in which they identify themselves using the old BIC).

- e) **Disable:** Disabling of a Participant forbids it from sending and receiving any messages. Thus, the IPS marks as rejected the messages sent by a disabled Participant and returns a response in pacs.002 format. Also, payment messages for a disabled Participant are marked as REJECTED. Users of the Participant still have access to the system through the graphical interface. Disabling a participant will forbid all IPS/Proxy messages to/from that participant. Blocking an account will only impact transactions to/from that account (pacs.008, pacs.004, pacs.009) based on the option: blocked, blocked for credit, blocked for debit.

The positions accumulated in the Participant's technical accounts can still be included in manual liquidity adjustment operations in order to transfer the funds back into RTGS, prior to removing the Participant.

The disable operation can be run at any time for a Participant in the ACTIVE state and it takes effect at the time of approval. If at the time a Participant is DISABLED, there are messages or payments in the process of processing that belong to the Participant, they will still be processed until they have final status. In other words, disabling a Participant only affects the messages received by the IPS at a later time (after approval) of the Participant's disabling.

For better granularity technical accounts can also be blocked/unblocked/block-incoming/block-outgoing. In the account menu there is a dropdown for account status with the following options: active, blocked for credit, blocked for debit, blocked for credit and debit. Either one can be selected for a specific account.

The Disable operation for participant will also automatically disable all users of that participant. The IPS Operator may re-activate users of that disabled participant. If the participant status is DISABLED, only the liquidity adjustment to the RTGS account can be performed.

- f) **Activate:** This is the reverse disable operation that returns the Participant (only participant not its' users) from the DISABLED state to ACTIVE. Only the IPS Operator users can activate a Participant.
- g) **Remove:** Performing the **Remove** operation can be performed under the following conditions: the IPS Operator users can remove a Participant. This operation requires an approval step. The Participant can be removed only if the following conditions are met:
 - a. The Participant is in the DISABLED state.
 - b. The Participant has an account position equal to zero (current balance = available balance = pending credit = pending debit = 0). In other words, the remove operation can only be performed after the Participant has been disabled and at least one

liquidity adjustment has been executed to clear off the obligations arising from the funding process prior to the blocking/disabling operation. If the participant is disabled then new transactions to/from that participant cannot be executed, so there is no risk of liquidity changing in IPS. After disabling, if the account balance in IPS is different than 0 then a defunding operation should be performed to move the funds out of IPS. Afterwards the participant can be removed. Verification of the remove operation conditions is performed both when initiating the removed operation and when approving it. The condition that the participant has an account balance of zero is checked both when the first user initiates the removal and at the moment when the second user approves the operation, to avoid inconsistency caused by modifications to the participant account while waiting for the approval of the removal.

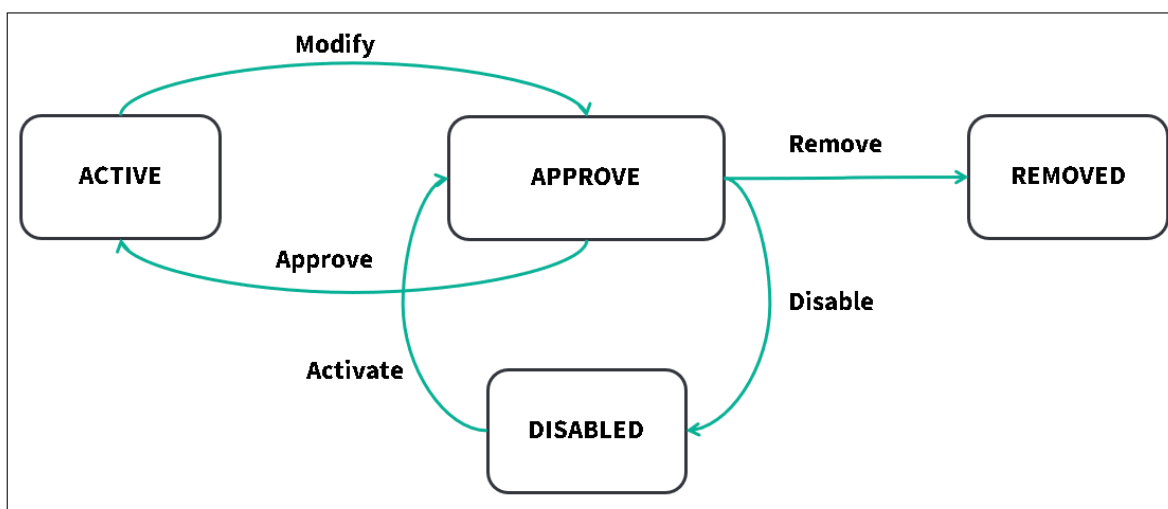


FIGURE 3. DISABLE/ACTIVATE PARTICIPANT STATUS TRANSITION

Note: In the case of the **Modify** Participant operation, it is not allowed to modify the following fields: b. Participant's group.

Online/Offline State

In addition to the Participant's main status, the system also monitors the status of a Participant's real-time connection to the system.

The IPS <-> Participant API integration is done on a Polling call arrangement. The participant advertises its availability by calling the API with a defined frequency. This ensures the connectivity is validated nonstop. The interval of the verification is configurable, if no call is done by the Participant, it will be marked as OFFLINE.

For efficiency, the responses by the IPS are delayed when no data is available, with the delay broken if data arrives. When the check interval is exceeded, the participant is marked as OFFLINE, this also causes the validation of payments toward him to fail with a specific error code. All the transitions

ONLINE/OFFLINE will cause an alert to be produced. The gaps in availability are recorded in a report as well.

Therefore, a Participant may be:

- **ONLINE**, which means that the STP application of the Participant is connected to the central system and is available to receive, send and process messages.
- **OFFLINE**, status indicating that the Participant is not available for receiving and processing messages. When validating a message addressed to an Offline Participant, IPS rejects the message with a specific error code. An offline Participant can send messages to the system.

The MMC monitors the status of the Participants' connection by comparing the time elapsed since the last call of the STP function to receive messages with the Participant Connection Timeout system parameter. The monitoring of the Participants' connection to the system can be done by the operators or users in the MMC and via API, participants.001.xsd.

Participant's ONLINE/OFFLINE status changes are not registered in the Participant audit because they may also occur as a result of some transient communication problems, which can generate a large number of records in a short time. Thus, the ONLINE/OFFLINE state is a (technical) state distinct from a business's status of a Participant and is largely similar to the rest of the entities: ACTIVE, DISABLED, REMOVED.

Types of Settlement

In terms of settlement processes, the system supports two types of settlement:

1. **Direct Settlement Participant (DP)** – Participant having own account in the settlement system (RTGS).
2. **Indirect Settlement Participant (IP)** – Participant for which the settlement is performed indirectly through the account of a settlement agent. The Settlement Agent must be a Direct Participant.

The process of changing the settlement type for a participant, only allowed for the IPS Operator users, is the following:

- From Direct Settlement Participant to Indirect Settlement Participant:
 - The system validates that the Participant subject to the change is not a settlement agent for other Indirect Participants. If there is an associated IP, the operation is not allowed.
 - After the change, the IPS Operator's users or users of the settlement Participant will configure the clearing limit of the Indirect Participant, which will enable him to make payments.
- From Indirect Settlement Participant to Direct Settlement Participant:

- No additional validation is required. After the change, the IPS Operator's users or users of the direct Participant will be able to configure the account limits.

Accounts

An IPS technical account will be generated automatically for each participant after they are mapped to a payment schema. Only after a participant is mapped to a payment schema, IPS is able to identify the processing rules for transactions and messages and the supported currencies. If a participant supports a specific currency it will be mapped to a payment schema defined with that currency and then IPS generates an account for that participant and currency. The mapping between participant and payment schema is done by the Operator.

IPS technical accounts are also generated for indirect clearing and/or settlement participants. Direct settlement participants will manage the account limits of their indirect settlement participants.

The following details are displayed for each account:

- Number – account identification which follows the pattern: participant BIC + Currency
- Account status – Active by default; can be set to Blocked, Blocked for credit, Blocked for debit, Removed.
- Currency
- Allow auto adjust liquidity – flag to indicate whether this account will be included in automatic liquidity adjustment events; in case of indirect settlement participants then only their direct settlement participant can change this option
- Low Limit – minimum balance for which the account is not included in the liquidity adjustment event
- Base balance – target balance for the liquidity adjustment event
- High Limit - maximum balance for which the account is not included in the liquidity adjustment event
- Before Holiday Limits increment – optional; This increment will be taken into account for the last event scheduled before one or more RTGS holiday days. All the limits (low, base, high) will be incremented with this value multiplied by the number of holiday days.

The IPS addresses the following types of accounts:

- IPS technical account – this is the account of an IPS Participant (any type of Participant), created and managed in IPS, calculates and manages the Participant's balances.
- IPS Subaccount – an IPS Technical Account belonging to an IPS Indirect Settlement Participant. The limits of these accounts are managed by users of their direct settlement participant.

- IPS MPFA - The Mirror Prefunded Account is an IPS Technical Account managed automatically by IPS. It is used to track the total amount of liquidity that participants have in IPS and will always be the opposite amount of the one in RTGS called the IPS pool account in RTGS.

The IPS Pool account is the account owned by IPS in the RTGS used for funding and defunding operations. Every transaction in RTGS between RTGS participant accounts and the IPS pool account is mirrored in IPS between the MPFA and direct/indirect settlement participant technical accounts.

Direct / Indirect Participant Interaction

Direct Settlement Participants can see information about their mapped indirect settlement participants related to RTGS communication: messages and reports for liquidity adjustment and billing. Direct Settlement Participants can manage the limits of their indirect settlement participants, can initiate manual funding/defunding for them and also decide whether they participate in the automatic liquidity process. Indirect Settlement Participants can see all these configurations but cannot modify them.

Direct Clearing Participants can see information about their mapped indirect clearing participants related to IPS API communication: messages, transactions, and reports about messages and transactions (reconciliation report, processing exceptions report).

If the indirect clearing participant moves to another direct clearing participant, the indirect clearing participant will see their own old messages/transactions but the new direct will only see the messages/transactions made after the switch.

3.1.2. User Groups

Each system user, from the IPS Operator or from the Participant, is assigned to a user group. There are two types of groups:

1. ADMIN, for the IPS Operator's users.
2. PARTICIPANT, for Participants' users.

The ADMIN type is created when the database is initialized, while the initial users are created.

PARTICIPANT type groups are created automatically when defining Participants during the approval step of the Participant create operation. Once the IPS Operator user has finished defining a new Participant and creating the associated group by default, the IPS Operator user can add users to this group.

The users from Participants are assigned to their own type of group, which has restricted access to data and IPS functionalities. If a user that belongs to group ADMIN can see the menus for create, modify, approve operations, a user that belongs to group Participant will only be able to see and access the operation List for most functionalities.

3.1.3. User Access Profiles

IPS allows the definition of custom profiles for each set of users with similar access rights. A profile contains the list of functions available in the system menu for users. The default **List**, **Create**, **Modify**, **Remove**, and **Approve** operations are applicable. For security reasons, all profiles are divided into two main types¹:

1. for Administrator, named ADMIN (in IPS and proxy module).

for Participants, named GROUP (in IPS and proxy module), SUPERUSER (in IPS and proxy module), AUTHORIZED (only for proxy) Groups are used for the purpose of data access control (participant users can only see own data while operator users can see all data) while profiles are used for role-based access control (to define what functions in the menu the users belonging to that profile can access).

From the default Participant profiles, SUPERUSER is the profile with maximal rights and is allowed to create other users, profiles and to interact with dispute management module.

The list of functions allowed by both types of profiles is presented in detail in the System Menu Annex.

The system automatically creates a default profile for each type described above when they are not detected at initialization. These profiles automatically created in IPS can be used as templates to define other specialized profiles but can also be used as such. The specialization involves restricting some rights from the total set allowed by the profile type.

A user may belong to only one profile that will define the list of functions that the user is authorized to execute but can be modified and associated to another profile by the administrator. When a user's profile is changed and the user is logged in, the user will be logged out to force a new user session which considers the updated profile.

Operations with Profiles

The following operations are allowed in the IPS:

- **List:**
 - the IPS Operator users can filter all existing profiles in the system. It includes Administrator and Participant profiles.
 - Participants' users can filter only Participant profiles. They can filter the general participant profiles and the profiles which belong to their institution.
- **Create:**

¹ This division makes it impossible to assign administrator functions to bank users by mistake.

- the IPS Operator users can create a new profile in the system. They can create Administrator and general Participant profiles.
- Participants' users can create a new profile in the system. They can create only a Participant profile which will belong only to their institution. Those profiles can't be assigned to the users of a different Participant.
- **Modify:**
 - The template profiles can't be modified by users. If it's required, the system will modify automatically the templates.

The default profiles contain the maximal rights allowed for each type of entity (Operator or Participant). New profiles can be created based on them at any time, but with fewer rights. In practice, these profiles are reviewed during UAT and rarely change, if ever. Therefore, updating these profiles to add more rights will be done as an application release but will not need a change request.
 - IPS Operator users can modify an existing profile in the system. They can modify Administrator and general Participant profiles.
 - Some participants' users (of profile type superuser) can modify a profile in the system, depending on their assigned profile. They can modify only a Participant profile which will belong only to their institution.
- **Remove:**
 - The template profiles can't be removed by users.
 - IPS Operator users can remove profile in the system. They can create Administrator, Viewer and general Participant profiles.
 - Participants' users can remove a profile in the system. They can remove only a Participant profile which will belong only to their institution.
 - At least a template for Operator users and a template for Participant users must exist always. These templates must exist in the system because all other templates are defined starting from them. It is not necessary to assign specific users to them. Those templates can't be removed but the IPS Operator can decide exactly what they should contain. After go-live, IPS Operator users can create specific templates and assign them to users as they see fit.
- **Approve:**
 - IPS Operator users can approve profile in the system. They can approve Administrator and general Participant profiles.

3.1.4. Users

IPS allows the definition of users who are intended to access the MMC administrative tool. The IPS stores in its own database the users. In addition to the default **List**, **Create**, **Modify**, **Remove**, and **Approve** operations applicable to all entities managed by the system, Users also support the following operations: **disable** and **activate**.

The IPS supports the following operations for managing users and operators accessing the graphical interface:

1. View
2. Add
3. Modify
4. Disable and activate
5. Remove
6. Approve

The flow of operations to add a user is:

1. Select the user group (Participant group or the IPS Operator – filtered according to logged-in user).
2. Fill in the user details and complete the registration after validation.
3. Another authorized user (operator or Participant depending on who initiated the change) approves the Add operation.

The information required to add a user is:

NAME/DESCRIPTION	FORMAT
Username	Text, max 100
Full name	Text, max 100
Password validity period (days)	Numeric, as defined in the security policy (default value of "Password validity period" parameter)
Maximum number of password attempts	Numeric, max as defined in the security policy (default value of "Maximum retry count" parameter)
Preferred language – for the default display of the user interface	Combo Box type control, with values: English (by default)
Group name	Read-only control, the group is preselected at input and cannot be modified later.
Profile name – User access profile	Combo Box type control, with the list of profiles that can be associated with the user. It depends on the group type (Participant or Admin).

Users can only be created by other users who have an administrator or superuser profile (or a profile derived from these).

The **Disable** operation temporarily restricts user access to the system. The user cannot be connected to IPS, and if the user is logged on when the disabling operation is approved, the current session is

terminated. A user may be disabled by an IPS administrator or automatically by the system if the number of consecutive incorrect password input attempts is equal to the parameter in the user profile above.

The **Activate** operation allows the administrator to reactivate a previously disabled user. If the current number of password input retries is equal to the maximum number allowed for the user, then it resets to zero.

The **Remove** operation marks the definitive and irrevocable ban of a user's access to the system. It can apply to any user.

User Identification Details

The IPS allows the IPS Operator (or Participant with user creation rights) to record the following information for each user:

NAME / DESCRIPTION	FORMAT
Username	Text, max 100
Full name	Text, max 100
Password	Text
Verify Password	Text
Password validity period (days)	Numeric, as defined in the security policy (default value of "Password validity period" parameter)
Maximum number of password attempts	Numeric, max as defined in the security policy (default value of "Maximum retry count" parameter)
Preferred language – for the default display of the user interface	Combo Box type control, with values: English, Georgian
E-mail address for receiving automatically generated reports	Text, max 100, e-mail address formatting validation, optional
Phone number, For receiving SMS alerts	Text, phone number formatting validation, optional
Group name	Read-only control, the group is preselected at input and cannot be modified later
Profile name – User access profile	Combo Box type control, with the list of profiles that can be associated with the user. It depends on the group type (Participant or Admin)

Operations with Users

The following operations are allowed in the IPS:

- **List:**
 - IPS Operator users can filter all existing users in the system.

- Removed users (using the MMC) are still visible alongside their history, displayed with status REMOVED.
- Participants' users can filter only users who belong to their institution.
- **Create:**
 - IPS Operator users can create a new user in the system. They can create users for all Participants and for System Operator.
 - Participants' users can create a new user in the system. They can create users who will belong only to their institution and assign profiles for them. User creation process is the same for all types of participants (direct/indirect, clearing/settlement), the difference is when configuring the Participant entity the specific flags must be set.
- **Modify:**
 - IPS Operator users can modify the details of any user in the system.
 - Participants' users can modify a user in the system. They can only modify a user which will belong only to their institution.
- **Remove:** Marks the definitive and irrevocable ban of a user's access to the system
 - The IPS Operator users can remove any user in the system.
 - Participants' users can remove a user in the system. They can only remove a user who will belong to their institution.
- **Disable:**
 - The Disable operation temporarily restricts user access to the system. He cannot be connected to IPS and if his is logged on when the disabling operation is approved, the current session is terminated. A user may be disabled by an IPS administrator or automatically by the system if the number of consecutive incorrect password input attempts is equal to the parameter in the user profile above.
- **Activate:**
 - Allows the reactivation of a previously disabled user. If the current number of password input retries is equal to the maximum number allowed for the user, then it resets to zero.
- **Approve:**
 - The IPS Operator users can approve any user in the system.
 - Participants' users can approve a user in the system. They can only approve a user who belongs to their institution.

User creation procedure

The operator entity (the dedicated group to which operator users are mapped), and a dedicated participant used to track liquidity adjustments (to mirror the prefunded account in RTGS) are created automatically on system initialization. The dedicated participant (owned by Operator) is initialized

and managed by IPS automatically, and its role is to facilitate tracking of liquidity operations: acts as a debtor for funding operations and creditor for defunding operations.

The system is initialized with 2 predefined users for the operator. They can be used to create all the other profiles and users and then to be deactivated for security reasons. The operator users are responsible for configuring Participants and for creating users for them. The usernames and passwords of these initial users are configurable (static parameters set at system setup) and on the first login they will be required to setup OTP for authentication. We do not recommend to keep these users after the initial setup (creating the actual users and profiles) since they are not bound to a specific person (which would be accountable for the actions performed in the system).

When creating a user the profile must be specified. The profile determines what access rights that specific user has. It is possible to change the assigned profile for users at any time. For example, participant users can be created with a profile that doesn't contain the "User – Create" option and then later be upgraded to a profile that contains that option.

3.1.5. Password Change

The system allows a user to change his password in accordance with the security policy rules set by the IPS Operator. The change is made in the user graphical interface. When the password has expired the user is redirected to the Password Change form and forced to change it.

If the user forgets its password then they can request a password from the login screen. They will be directed to enter their username and receive a token on the registered email for that user. Using that token, they will be able to set a new password and confirm it.

3.1.6. Calendar

The Montran IPS solution works on both the working days of the RTGS system and the rest of the calendar days, which is why IPS allows the definition and maintenance of two calendars: an automatically synchronized calendar with the RTGS settlement system and a generic calendar for all calendar days.

In both calendars, it is permitted to define a working day or non-working day for each settlement currency. The RTGS synchronized calendar must reflect the working days defined in the RTGS system, while the generic calendar is defined in such a way that the IPS is available 24/7/365-366.

The RTGS calendar contains the following types of day:

- Business day - working day in which a settlement window is defined for liquidity transfers between IPS and RTGS, and during which at IPS EOD the balance messages are sent
- Non-working business day – no settlement window, but balances are still sent at IPS EOD
- Holiday – no interaction between IPS and RTGS

The maintenance operations of the calendar are only available to operators, and consultation is available to users: by accessing the menu items Maintenance – Calendar – List (to see RTGS business

days), Maintenance – Timetable – List (to see thresholds such as daily rollover, settlement window start, settlement window stop) and Event Cycle – List and Monitoring – Event Scheduler (to see all liquidity adjustment events and their status). The operations available for calendar management are the standard ones: **View**, **Add**, **Modify**, and **Remove**. IPS will communicate with RTGS automatically (as described in Annex F) to obtain working days and hours and then will update the calendar accordingly. In case of communication failure an alert will be raised, and the calendar will be set manually by Operator user.

A calendar record corresponds to a calendar month, and it contains information about the type of calendar day of that month: working day or not. The IPS automatically creates a calendar entry at the beginning of a new month if the system administrator did not add that recording in advance. IPS creates the calendar by default for an year in advance. At the beginning of a new month, it creates the corresponding month for next year so that it always has at least 12 months registered. This record has a default configuration:

1. For the RTGS type only: Monday to Friday are working days. This is just the default setup. In that case, RTGS will push those updates to IPS which will update the calendar.
2. For the IPS type: all days are working days.

In the **Modify** operation, it is only allowed to change the status (from working to holiday and vice versa) of future days, while past days can no longer be modified.

Working days, holidays can still be updated for current month, for past month cannot. Modification will come into effect immediately after the approval step.

The **Remove** operation can only be performed for future months. Calendar records for the past months cannot be removed.

Operations with Calendar

The following operations are allowed in the IPS:

- **List:** IPS Operator users and Participant users can filter all existing calendars in the system.
- **Create:** Only IPS Operator users can create a new calendar in the system, alongside the automatic sync process from IPS. When creating a new calendar there is a dropdown for type IPS/RTGS. The month, year, and currency are mandatory fields. IPS doesn't allow two calendars with the same month, year, and currency. This operation is allowed only for future months.
- **Approve:** Only IPS Operator users can approve the modifications performed over a calendar entity.

3.1.7. Daily Operation Timetable

All timetable events are set by IPS Operator users and can be run automatically or manually. The two types of calendars have different events. The allowed operations for a timetable are **List**, **Modify**, and **Approve**.

The timetable events of the RTGS synchronized calendar are described in the table below.

EVENT – RTGS SYNCHRONIZED CALENDAR	DESCRIPTION
Change of working day (roll-over)	Closing time of the current working day and the beginning of the next working day. During this operation the system takes Participants' balances at the end of the current day and saves them as initial balances for the next day.
The beginning of the settlement period	The moment of a business day when the IPS starts sweeping operations. The actual event times are specified by the mechanism defined in the Settlement Timetable chapter. This moment must correspond to the time from which the RTGS system can receive and process liquidity adjustment messages.
The end of settlement period	The moment of a business day, marking the time from which IPS will no longer perform sweeping operations. This time must correspond to the closing time of the RTGS activity settlement messages.
RTGS End of Day	The moment when RTGS executes EOD event. It is used to track if reconciliation message is received correctly from RTGS.

The calendar for non-working days in the RTGS system has one event: change of working day that will take effect on the next calendar day. This operation resets the internal sequences of the incoming and generated messages, as well as the transfer of current day balances as opening balances for the next day. The processing of payment instructions is not interrupted during the execution of this operation.

The IPS allows the IPS Operator to define these timetable events in a centralized manner, similar to the Daily Timetable, having the possibility to define these timetable events in the form of a template that will repeat in the following working days but also in the current form that will be valid only for the current working day.

The IPS timetable events for settlement window start and settlement window stop are at the same times as the ones for RTGS. A mechanism to synchronize automatically with the RTGS timetable and set the times in the default IPS timetable as these are received from RTGS will be added according to Annex F.

Operations with Timetable

The following operations are allowed in the IPS:

- **List:** All users can filter the existing information of the timetable events.
- **Modify:** IPS Operator users can modify the timetable. The changes allowed are:
 - The execution of an event is manual or automatic.
 - In the case of automatic execution, the time when the event should be executed.

→ There are two types of timetables that can be modified:

- Default: Timetable that represents the current business day.
- CCY-RTGS: Timetable that represents the RTGS events.
- **Approve:** IPS Operator users can approve the changes performed over a timetable
- **Execute:** IPS Operator users can execute manually timetable events which are configured with manual schedule.

Manual intervention on the timetable will be possible in case of not receiving data from RTGS.

There is a contingency menu for closing RTGS flows manually. There is also the functionality to re-send the messages to RTGS on the decision of the operator user. The daily rollover event, which also performs the changing of the business day, can be set to manual mode in case of contingency.

Payment Schemas Parameters

The Payment Schema defines a set of rules for different businesses defined by the IPS Operator using message codes (local instrument and category purpose) and custom time deadlines. Multiple Payment Schemas can be mapped to the same Participant allowing it to participate in multiple businesses defined by the IPS Operator.

The set of parameters for a payment schema are listed in the table below. The default **List**, **Create**, **Modify**, **Remove**, and **Approve** operations are applicable. Only IPS Operator users can perform the operations.

NAME/DESCRIPTION	FORMAT
Name	Text, max 20
Currency	Combo Box type control with currencies defined in the system.
Local Instrument	Combo box control with local instrument codes defined in the system. Default All
Category Purpose	Combo box control with category purpose codes defined in the system. Default All
The maximum amount allowed for a payment instruction.	Numeric, between 0.01 and 999999999999.99.
The maximum business day range from the processing of a SCT instruction in which a Recall instruction is allowed. This parameter is not validated by the system against the credit, it uses the declared OrgnlIntrBkSttlmDt (due to data retention and performance concerns).	Numeric, between 0 and 99.

NAME/DESCRIPTION	FORMAT
The maximum business day range from the processing of a SCT INST instruction in which a Recall instruction is allowed for duplicate reason. This parameter is not validated by the system against the credit, it uses the declared OrgnlIntrBkSttlmDt (due to data retention and performance concerns).	Numeric, between 0 and 999
The maximum business day range from the processing of a SCT INST instruction in which a Recall instruction is allowed for fraud reason. This parameter is not validated by the system against the credit, it uses the declared OrgnlIntrBkSttlmDt (due to data retention and performance concerns).	Numeric, between 0 and 999
The maximum business day interval measured since receiving a SCT INST instruction/ Recall instruction where it is allowed to send a Recall Response (Positive or Negative) instruction. This parameter is not verified by the system, because Recall Response processing does not perform Recall instruction identification, it uses original IntrBkSttlmDt (due to data retention and performance concerns).	Numeric, between 0 and 99.
Timeout Deadline – the maximum processing time in seconds of a message in the IPS. If this time is exceeded, the payment is cancelled. This is the total processing time since the sender's initiation of the payment (adding the timestamp), includes the internal processing of the IPS, the processing by the receiving Participant and the transmission times of the messages (from the initiating Participant to the IPS, from the IPS to the recipient and from recipient to IPS). The transmission time of the end flow messages from IPS to initiator and recipient Participants are not included because, as final messages, they can no longer affect the transaction. The deadline is calculated against the following XML field: FIToFICstmrCdtTrf/CdtTrfTxInf /AcceptncDtTm, which is under the control of the sending participant. This parameter is validated for pacs.008 and pain.001.	Numeric, between 1 and 999.
Initiation Deadline – the maximum time in seconds for a payment/message initiation from a participant. This is the total time passed from the moment of the sender participant's initiation of the payment/message (adding the timestamp) until being received by the IPS. This check is done to allow enough time for the receiver to process the payment/message within the SLA. For example if sender participant builds the message but only sends it after 18 seconds, then considering a deadline of 20 seconds the receiver would have at most 2 seconds (minus network and IPS processing time) to validate and accept the payment, and this is the situation which we aim to prevent with this parameter. The deadline is calculated against the following XML field: FIToFICstmrCdtTrf/CdtTrfTxInf /AcceptncDtTm, which is under the control of the sending participant.	Numeric, between 1 and 999

NAME/DESCRIPTION	FORMAT
This parameter is validated for pacs.008, pacs.004, pain.001 and pain.013.	
Allow Request to Pay (RTP) – if the Participant mapped to this schema is allowed to send RTP messages.	Checkbox
Slow RTP Deadline – the number of days after which the IPS will automatically close the RTP if it hasn't been closed by a positive/negative answer from the beneficiary customer.	Numeric, between 1 and 90
Fast (Merchant) RTP timeout – number of seconds after which the IPS will automatically close the RTP if it hasn't been closed by a positive/negative answer from the beneficiary customer. This timeout applies to the Merchant initiated RTPs.	Numeric, between 1 and 999
Description Free text field to add details about the purpose of the payment schema	Text, optional, 500 characters

3.1.8. STP Digital Certificate Management

The IPS solution allows IPS Operator's users and Participants' users to import the digital certificates used by the Participants into the system database. The import of certificates is performed in the MMC using the graphical interface. If the function is used by the user of a Participant, then the system allows only the import of the certificates of that Participant. IPS Operator's operators can upload certificates for any Participant defined in the system.

Digital certificates uploaded to the system are used for two purposes:

- Authentication of the Participants in IPS and PROXY.
- Verification of the digital signature applied to the messages by the Participant for IPS and PROXY.

When uploading a certificate to the system, the operator/user needs to specify the following information:

- The Participant to whom the certificate belongs.
- The purpose of using the certificate:
 - Authentication, or
 - Digital signature.
- The file that contains the X.509 certificate in PEM format.

The system extracts from the certificate and displays the following information (additional fields) for operators:

- Name of the issuer of the certificate (Issuer Name)
- Name of the certificate holder (Subject)
- Serial number of the certificate (unique for an issuer)
- Validity period (From, Up to)
- Certificate status.

A Participant may have multiple digital certificates registered in the system, but a digital certificate can be imported only once into the system and is always associated with a single Participant. The uniqueness of the certificate is validated on the basis of the issuing entity (Certificate Authority – CA) and the serial number of the certificate.

The certificates for authentication and digital signatures can be the same, provided the usage flags allow it – typically they are distinct, but this is not enforced.

Upon uploading a certificate, the system also validates the certificate validity period information so that a certificate can only be used during the validity period. After expiration, the certificate, although active, cannot be used to authenticate connections and validate later messages.

The certificates will be produced by IPS Operator’s B-trust CA. For user certificates renewal is permitted with the same serial number. STP certificates need to be issued again.

In order to cover the security requirements, the IPS must have access in some way to participant certificates (the public keys) in order to perform the necessary validations. In the current implementation the system allows for uploading the certificates, either by system operator users or participant users, while also mapping the use case (digital signature, TLS communication, proxy service DS, proxy service TLS).

Operations with Digital Certificates

The following operations are allowed in the IPS:

- **List:**
 - IPS Operator users can filter all existing digital certificates in the system.
 - Participants’ users can filter only digital certificates who belongs to their institution.
- **Create:**
 - IPS Operator users can create a new digital certificate in the system. They can create digital certificates for all Participants.
 - Participants’ users can create a new digital certificate in the system. They can create digital certificates which will belong only to their institution.
- **Disable**

- A digital certificate may be temporarily disabled for use by using the **Disable** operation. It can be reactivated later by the **Activate** operation. The purpose of the Disable operation is to have a temporary ban on using a digital certificate, as opposed to the **Revocation** operation made by the issuer of the certificate that is irreversible. Disable can be used, for example, in case of unconfirmed suspicions of compromising the security of a digital certificate.
- IPS Operator users can disable a digital certificate in the system. They can disable digital certificates for all Participants.
- Participants' users can disable a digital certificate in the system. They can disable digital certificates which belong only to their institution.
- **Activate:**
 - Allows the reactivation of a previously disabled digital certificate.
 - IPS Operator users can activate a digital certificate in the system. They can activate digital certificates for all Participants.
 - Participants' users can activate a digital certificate in the system. They can activate digital certificates which belong only to their institution.
- **Remove:**
 - IPS allows the removal of a digital certificate which was previously disabled. Because removing a certificate is an irreversible operation, its primary purpose is to undo the effects of registering a wrong certificate (or a correct certificate associated with a wrong Participant) in the system.
 - IPS Operator users can remove a digital certificate in the system. They can remove digital certificates for all Participants.
 - Participants' users can remove a digital certificate in the system. They can remove digital certificates which belong only to their institution.
- **Approve:**
 - IPS Operator users can approve any digital certificate in the system.
 - Participants' users can approve a digital certificate in the system. They can only approve a digital certificate who belongs to their institution.

A digital certificate may be temporarily disabled for use by using the **Disable** operation. It can be reactivated later by the **Activate** operation. The purpose of the **Disable** operation is to have a temporary ban on using a digital certificate, as opposed to the **Revocation** operation made by the issuer of the certificate that is irreversible. Disable can be used, for example, in case of unconfirmed suspicions of compromising the security of a digital certificate.

A **disabled** Certificate can be removed later.

Removing a certificate is an irreversible operation. Its primary purpose is to undo the effects of registering a wrong certificate (or a correct certificate associated with the wrong Participant) in the system.

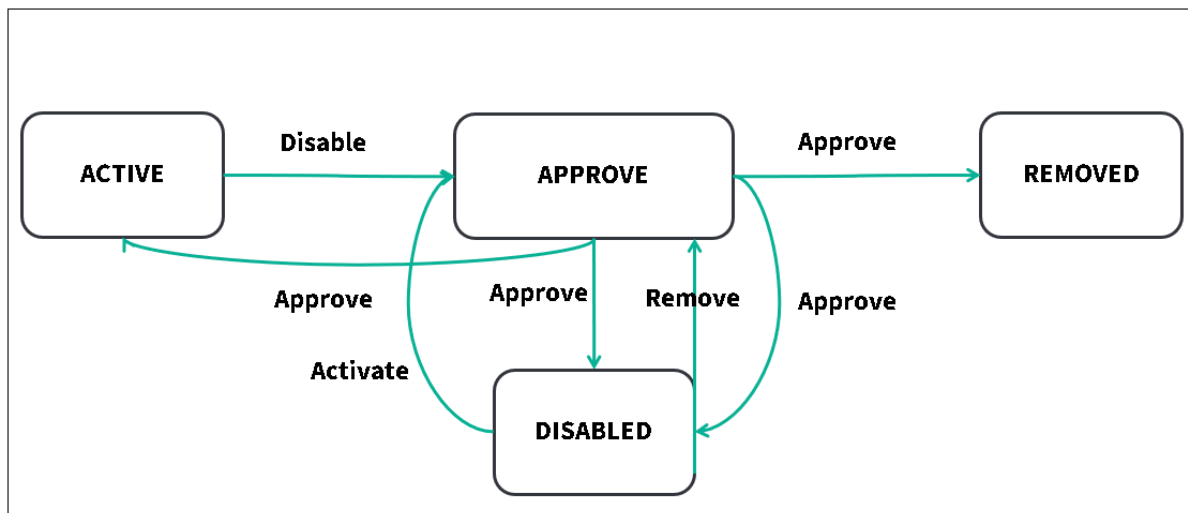


FIGURE 4. STATUS TRANSITION CHART FOR A DIGITAL CERTIFICATE

In addition to the **Disable** operation, which is reversible, a certificate may also be revoked by the issuing authority (CA). Following the revocation of a certificate, which is an irreversible operation, the issuing authority issues an update to the revoked certificate list (CRL).

As part of the digital certificate verification process, IPS also considers the list of revoked certificates.

All certificates, regardless of their status (VALID, EXPIRED, REVOKED, REMOVED) are stored indefinitely in the system; they can still be removed during the archiving procedure. Thus, these certificates can be used for further investigations (manual procedures), if required.

The IPS solution will automatically update the status of imported digital certificates following periodic expiration and revocation checks.

- The IPS solution will automatically check the validity of each certificate in ACTIVE or DISABLEED status. If the validity of a certificate has expired, the system will change its status to EXPIRED.
- When processing a new CRL file from the disk, the system will check the list of revoked certificates from CRL with existing ones in the system in ACTIVE or DISABLEED state. If such a revoked certificate is detected, it is marked with REVOKED status, an irreversible state in the system.

3.1.9. Messages

All messages processed by IPS, including messages exchanged with RTGS using message queues and messages exchanged between IPS participants through the REST API, can be listed and filtered in MMC. Participants can see their own messages.

A dedicated menu exists for messages exchanged with RTGS called “Interface Messages”. This functionality allows listing and filtering these messages and also manually inserting messages, as a contingency measure. Manual injection of messages is a 4-eyes check operation. After approve, IPS will process that message as if it was received on the IPS Input queue.

3.1.10. Four-eyes Check mechanism

All management operations in MMC are done using the Four-eyes Check mechanism. Every Create/Modify/Remove operation must also be approved by a different user before it is considered final. In case of Reject the operation is cancelled with no effect in IPS.

Example of “Approve” screen for an entity in the system (User) is displayed below:

Maintenance / User / Approve (USRAPP002)
Result 1 of 1

User
Approve
Active

Please review the information before submitting

User

Username *

test

Full Name *

other name

Email Address

Not specified

Password

Password

Verify Password

Password validity period (days) *

90

Maximum password retry count *

3

Preferred Language

English

Group Name

RTP

Profile Name

ADMIN

Discard
Reject
Approve

^ Audit
Audit Details

Operation	User	Time
Modify	admin1	20241218 14:38:40.729
Approve	admin1	20241218 14:38:29.839
Enter	admin1	20241218 14:36:27.299

The user reviewing this operation can decide whether to Accept or Reject the operation and press the specific button: Approve/Reject. During this review the user can see additional details by scrolling down to the audit list on the same screen.

In this example, the last operation was a “Modify”, performed by User “admin1” at timestamp “20241218 14:38:40.729”. By clicking on the Audit Details button more details will be displayed about the entity:

Maintenance / User / Approve (USRADTLST001) ← Result 1 of 3 > ?

User Approve

Audit

User admin1 Time 20241218 14:38:40.729	Operation Modify
--	---------------------------------------

User

Username * test Email Address Not specified	Full Name * other name Previous: test test
---	---

Password

Password Password validity period (days) * 90	Verify Password Maximum password retry count * 3
--	---

Preferred Language English	Group Name RTP
Profile Name ADMIN	

In this example, the fields which were changed through the “Modify” operation are highlighted.

The operations which can be done without Four-eyes Check (approve from another user) are:

1. broadcast alerts, from Maintenance – Alerts – Broadcast
2. switch processing node leader site (used for emergency situations), from Monitoring – Node Status – Switch Site button
3. starting/stopping internal queues (used for emergency/maintenance situations), from Routing – Control

3.1.11. Merchant Category Codes

IPS allows the Operator to manage Merchant Category Codes through MMC user interface. The users are allowed to list/create/remove MCC and their groups, as well as the relationship between them. MCC management is subject to four-eyes check.

Besides the manual management of codes there is also the option to upload a file with the full list of MCC. In case of file upload then the entire list in MMC is replaced with the new codes.

3.1.12. Account Upload (Proxy)

This functionality enables Account Data Provider participants to upload files containing their customers' account information to the Proxy Module. The upload is performed through "Account Upload" menu in MMC. This enables the user to select a file containing the Account Data using the standard format (acmt.022 – described in Proxy Participant Interface Annex) and upload it into the system. The uploaded files will not have to be digitally signed, since they will be uploaded by MMC users and subject to 4 eye check mechanism. After Approve step, the changes are forwarded to the processing nodes to be validated and applied to the database.

The Proxy Module supports the following operations for Account File Upload:

1. View – list the previously uploaded files
2. Add – the operation to upload a new file
3. Approve – approval of the upload operation

Liquidity Adjustment

The liquidity adjustment process is configured through multiple entities in MMC: Accounts, Settlement Method, Events.

.Participants in IPS who participate in liquidity adjustment process will have accounts in RTGS which will participate in funding/defunding operations on IPS pool account in RTGS. On the other hand, IPS pool account from RTGS mirrors the MPFA in IPS.

3.2.1. Settlement Method

IPS allows the IPS Operator to configure the settlement method, with the following parameters:

4. Name – identifier for the settlement method, generated by default as "RTGS-GEL"
5. Settlement Identifier – optional parameter; if set then IPS will add this value to the dedicated TTC field on messages sent to RTGS (/CdtTrfTxInf/PmtTpInf/SvcLvl/Prtry XML field in pacs.009). The specification of the TTC codes is in the scope of the specific RTGS system.
6. Supported currencies – Fixed "GEL"
7. Institution Code – BIC of the RTGS system, set in messages sent from IPS to RTGS
8. Type – Fixed "RTGS-Liquidity Adjustment" – the identifier for the Prefunded settlement model

The following operations are allowed in the IPS for configuring the settlement mechanism:

1. **List:** IPS Operator users can list the existing settlement methods in the system.

2. **Modify:** IPS Operator users can modify the details of a settlement method.
3. **Approve:** IPS Operator users can approve the changes performed to a settlement method.

3.2.2. Schedule of RTGS Events

Liquidity adjustments can be performed between the timetable events: *settlement window start* and *settlement window stop* from the timetable, that is synchronized with the RTGS system timetable.

In the Maintenance - Account menu from the MMC, there is a flag/ Checkbox for each participant account whether to include the automatic liquidity adjustment events. If the checkbox is selected then the participant is included in the automatic events for liquidity adjustment.

The definition of these moments results in a series of events that will be executed during the working day and whose timeline and evolution can be consulted at any time through a monitoring screen.

The settlement timetable has the following features.

	FIELD NAME		DESCRIPTION
1	Name		Event name/window of events.
2	Event type		(i)Liquidity Adjustment (ii)IPS Balances to RTGS (with a conditional field for delay time between balance capture and sending)
3	Define type		Template / Current Template events cause the creation of current events for all working days defined in the RTGS calendar and can be executed in the interval defined in the RTGS timetable. Current events only exist in the current day.
4	Currency		The currency for which the event / set of events is defined.
5	Window/Discrete event		If the event is a window type, then it will generate several discrete events starting with one hour: minute to another hour: minute and with a defined periodicity.
	Discrete events	Window events	
6	Time		Moment of time hour: minute of the event: minute of the event.
7		Beginning window	Moment of time hour: minute for a succession of events.
8		End window	Moment of time hour: minute when the sequence of events ends.
9		Periodicity	Expressed in minutes. The length of time between two events in the window.

The IPS has a launch mechanism in place of various events that ensure the chronological execution of the events as they were defined. From minute-to-minute IPS checks if there is any event to launch.

Events eligible to be launched must meet the conditions:

- To have a **discrete event** defined at the current hour and minute or before the current time that was not initiated, **or**
- To have an **event window** to generate, depending on its periodicity,
- An **event** at current hour and minute or before the current time, which has not yet been executed.

After identifying such an event, an entry specific to the execution of this event is created in the database, which is set to the CREATED status and is executed according to the type of event.

The list of executing or ready-to-run events can be viewed in real-time on the Monitoring – Event Scheduler screen.

3.2.3. Prefunded Settlement Model

IPS technical accounts are prefunded via the RTGS, therefore the settlement between parties happens internally, one payment at a time. The total amount prefunded mirrors the IPS pool account in RTGS. In order to track liquidity adjustments, the system automatically creates an RTGS mirror account in IPS, called the Mirror Prefunded Account (MPFA). The balance of this account is always equal to the sum of all the Participant's IPS technical accounts but as a debit (IPS technical account will always have a balance more or equal to 0). This is also a mirrored view of the IPS account in RTGS. It is created in IPS to provide additional information about liquidity and to serve as creditor or debtor for liquidity adjustment operations.

Interaction with RTGS is only required:

- When a Participant's technical account available balance is below the low limit, IPS will submit to RTGS a message debiting the Participant settlement account and crediting its own (the IPS pool account in RTGS).
- When a Participant's technical account available balance exceeds the high limit defined, IPS will submit to RTGS a message debiting the IPS pool account in RTGS and crediting the Participant settlement account.
- When a Participant user initiates a manual fund or defund operation through the MMC, IPS will submit to RTGS a message
- When a Participant initiates funding directly through RTGS, a notification message is sent to IPS which will also create a transaction between the MPFA and Participant's IPS technical account

The limits (Upper and Lower) based on which IPS determines if a participant has sufficient funds or not, are calculated from the template defined in the Account (profile record) menu entry, which corresponds to the IPS technical account that is automatically opened for each participant and for the GEL currency. The limits used by the automated liquidity adjustment events are configured by each Direct settlement participant individually (except indirect participants, for which their direct settlement participants configures limits).

The following account limits are defined in the MMC account menu:

- Low Limit – if the available balance is less than this amount during the automatic execution of a Liquidity Adjustment event then a funding operation is initiated automatically
- Base Balance – target balance for the Liquidity Adjustment event
- High Limit – if the available balance is more than this amount during the automatic execution of a Liquidity Adjustment event then a defunding operation is initiated automatically

Besides the limits, the following parameters in the Account menu are also taken into account:

- Before Holiday Limits Increment – optional; This increment will be considered for the last event or event window scheduled before one or more RTGS holiday days. All the limits (low, base, high) will be incremented with this value multiplied by the number of holiday days.
- Allow Auto Adjust Liquidity – if this parameter is unchecked then this technical account is not included in the automatic Liquidity Adjustment events.

Based on these values a set of limits are calculated each day for the Liquidity Adjustment event executions. If the next day is an RTGS working day, then the limits for the current day are the same as the ones defined in the Account menu. If the next day is an RTGS holiday, then the limits for the current day and the target balance will be increased by the value of the “Before Holiday Limits Increment” (set by each participant) multiplied by the number of holiday days. These calculated limits can also be filtered and modified by participants from the “Account Limit” menu.

For example, if the following days are a weekend, then the liquidity adjustment event will use the following limits during execution: low + 2 * increment, base + 2 * increment, high + 2 * increment. Considering the set limits of a Participant: min-1000, base-2000, max-3000, increment 4000: for the liquidity adjustment event executions within the scheduled event window, prior to a period of two holidays, the following adjusted limits will be used: low: 9000 (1000+2x4000), base: 10000 (2000+2x4000), high: 11000 (3000+2x4000).

The interface uses the credit RTGS settlement messages (confirming that settlement was successful) to either debit or credit the respective participant accounts (those messages are defined in the Annex F – RTGS Interface).

Based on liquidity adjustment events or manual operations in IPS, messages are generated with the gross amount and sent to RTGS, one for each IPS participant. The system tracks all liquidity

adjustments and allows for at most one ongoing adjustment per participant, in order to avoid duplicating adjustments in case of slow response from RTGS.

When listing the transaction generated for a liquidity adjustment (Clearing – Transaction - List), if there is an exception message associated with that transaction that could not be delivered to RTGS, then this information will be presented as well, allowing the user to quickly identify and navigate to the exception message and take the appropriate action.

Automatic Liquidity Adjustment

IPS can achieve automatic liquidity adjustment through continuous execution of liquidity adjustment events. This is done by defining a “Liquidity Adjustment” event of type window with a short cycle time (i.e. once every minute, configurable parameter by the operator). This ensures that throughout the settlement window period whenever the account limits are breached for a participant’s IPS technical account, after a short time, they will be evaluated by the execution of an event from the window and funding/defunding operation will be initiated.

This setup of a window event with a short cycle time will have no impact on the system performance since the event execution and RTGS communication is done through the MMC component and interaction with PM (the component which handles transaction processing) is only needed to register the transactions in IPS associated with funding/defunding operations, and their final status.

Liquidity Adjustment Event Execution

During each execution of a Liquidity Adjustment event (there can be multiple executions for window events), IPS performs the following steps:

- It creates an internal entry attached to the event for each technical account included in the automatic liquidity adjustment process, to keep track of the changes, called “Liquidity Adjustment Request”
- Check the available balance of each technical account included in the automatic liquidity adjustment process:
 - If the balance is within the low and high limits then the Liquidity Adjustment Request for that account is considered complete, with status “NO_ACTIVITY”
 - If the balance is less than the low limit amount, then a funding operation is initialized by sending a message to RTGS and the Liquidity Adjustment Request is moved to status “WAIT_RESPONSE”. If this operation is performed for an indirect settlement participant in IPS then the RTGS message will indicate the direct settlement participant as the debtor. A transaction is also created in IPS, on status HOLD, with debtor MPFA and creditor the participant technical account.
 - If the balance is more than the high limit amount, then a defunding operation is initialized by sending a message to RTGS and the Liquidity Adjustment Request is moved to status “WAIT_RESPONSE”. If this operation is performed for an indirect settlement participant in IPS then the RTGS message will indicate the direct

settlement participant as the creditor. A transaction is also created in IPS, on status HOLD, with creditor MPFA and debtor the participant technical account.

- While the event is ongoing and there are still messages sent to RTGS with no response yet, the status of the “Liquidity Adjustment” event execution as seen in the Event Scheduler monitoring screen is “Partial Completed”
- Continuously check for updates from RTGS for the initialized credit transfers:
 - If RTGS replies with a positive status, then the transaction in IPS is moved to status COMPLETE. If the operation was a fund, then the amount will be added to the IPS technical account’s current and available balance. If the operation was a defund then the amount is subtracted from the IPS technical account’s current balance (it was not part of the available balance since it was on HOLD). The Liquidity Adjustment Request is moved to status SETTLED.
 - If RTGS replies with a negative status, then the transaction in IPS is moved to status CANCELLED. If the operation was a fund, then there is no impact on the IPS technical account balance. If the operation was a defund, then the amount is added back to the technical account’s available balance (since it was previously on HOLD). The Liquidity Adjustment Request is moved to status CANCELLED.
- After all Liquidity Adjustment Requests are finalized the status of the “Liquidity Adjustment” event execution is moved to status COMPLETED.

For example, if at the execution moment, the Available Balance of that Participant is: Available balance = 2500 then, considering the calculated limits of (8000, 9000, 10000), the following calculations are performed:

- $\text{balance} < \text{low?} \Rightarrow \text{Adjustment (funds requested from RTGS)} = \text{base (10000)} - \text{balance (2500)} = 7500.$
- $\text{Balance} > \text{high?} \Rightarrow \text{not the case in this particular example}$

The rest of the flow is as described in this section: a funding operation for 7500 is initiated and sent to RTGS, IPS waits for response from RTGS, and the statuses are updated accordingly at each step.

When initiating a funding or defunding operation from MMC, IPS creates a transaction that marks the initiation of this process in the PM as well. The purpose of this transaction is to move the funds to/from the MPFA defined in IPS from/to the IPS technical account of the indicated party. The status of this internal transaction is HOLD. The funds which are in HOLD status (combined from liquidity adjustment operations and IPS transactions) are visible in MMC on the account monitoring screen. The amount which is in HOLD cannot be used to initiate transactions, therefore it guarantees that the funds used in IPS transactions while a liquidity adjustment operation is active will not cause the balance of the IPS technical account to go below 0. Therefore, IPS ensures uninterrupted flow of payments between Participants during the sweeping procedures, amounts are bound to obey the available balances.

Multiple Liquidity Adjustment event executions can be ongoing at the same time, so there is no risk that one execution waiting for the response from RTGS for a Liquidity Adjustment Request will affect other subsequent event executions. For example, if the limits for a Participant A are breached while the previous event execution is still waiting for the status of a fund/defund operation for Participant B, then for Participant A a new fund/defund operation will be generated during the next execution of the event, without waiting for the update on Participant B.

In case of holiday as next calendar day, the same event window will be used (according to our proposed configuration of an event window spanning the whole settlement window interval) and the limits for the calculation step will be the ones that entail the increase required for the holidays, calculated automatically by the IPS. Therefore, the updated limits for the holidays will be taken into account for all the adjustments executed in that day.

Manual Funding/Release of Funds

IPS offers the possibility for the participants to manually fund or release funds in the IPS accounts during the RTGS working hours. The process is initiated via the Participant Front-end via the MMC module and the result is a specific RTGS message that is generated by IPS along with an IPS transaction initially posted in HOLD status. When the response is received from the RTGS the internal transaction is finalized and posted accordingly. Manual operation can be performed any time during the settlement window, even if the automatic liquidity events are being executed at the same time.

Funding/Release of Funds for Indirect Participants

Indirect participants are IPS participants that do not participate in RTGS, so they are unable to fund their IPS technical accounts themselves. Indirect participants are registered in IPS along with their direct participants, which will support their funding operations.

Manual funding of an Indirect Participant is performed by its Direct Participant using its funds in IPS. The operation is posted immediately, and the Indirect Participant may use the received funds without any other limitation.

Only the Direct Settlement Participant can decide whether its Indirect Settlement Participants can participate or not in automatic liquidity adjustments, and only the Direct Settlement Participant will fund/defund manually its Indirect Settlement Participant IPS technical account. Both the Direct Settlement Participant and Indirect Settlement Participants will have a technical account in IPS.

Automatic liquidity Adjustment for Indirect Participants is also available in IPS based on the limits defined for the Indirect Participant's accounts. The RTGS message is sent to the Direct Participant mentioning the Indirect Participant identifier. The RTGS may respond positively or negatively to the request, and if the response is positive, the IPS will consider that the incoming funds are for the Indirect Participant. In IPS, the amount is moved directly to the indirect settlement participant technical account.

Message and Payment Processing

3.3.1. Sending, Receiving, and Processing Messages between IPS and Participants

In the IPS solution, payment processing is carried out by the PM. The messages supported by IPS are listed below.

Payment messages are sent and received individually, one message at a time, through the communication mechanism implemented by the PM.

1. The SCTInst payment schema messages, as defined in the SEPA Instant Credit Transfer Interbank Implementation Guidelines v 1.0. and Request to Pay (S RTP) scheme rulebook :
 - a. pacs.008.001.012 – FI to FI Customer Credit Transfer
 - b. Pacs.009.001.11 – FI to FI Credit Transfer
 - c. pacs.002.001.014 – Positive and Negative Confirmation Message.
 - d. camt.056.001.11 – Recall of an SCT Inst.
 - e. camt.029.001.13 – Negative Response to a Recall of an SCT Inst and to Request for Cancellation in case of camt.055 for pain.013
 - f. pacs.004.001.13 – Positive Response to a Recall of an SCT Inst.
 - g. pacs.028.001.06 – SCT Inst Transaction Status Investigation.
 - h. pain.013.001.11 – Request to pay, supporting debit transaction flow, purely informational, passed through the system to the debit party, to be followed by a pacs.008 and pain.014 (positive scenario) or pain.014 only (negative scenario).
 - i. pain.014.001.11 - Request to pay response, used when the debit requested via previous pain.013 is accepted or refused.
 - j. pain.001.001.012 – Payment Initiation
 - k. pain.002.001.010 – Payment Initiation Status Report
 - l. camt.055.001.012 - Request for Cancellation (RfC) or Request for Status Update of a RfC
2. Messages supported by the IPS solution that are not part of the SCT Inst:
 - a. positions.001 – The IPS response message to the Participant with information on the balance of technical accounts, the available balance.
 - b. Participants.001 – Participants List Table
 - c. camt.053.001.13 –Reconciliation Message

3. Messages supported by IPS in relation to RTGS, part of ISO20022 standard:

- a. pacs.009.001.011 – Credit Transfer
- b. pacs.002.001.013 – Status Report
- c. pacs.029.001.002 – Multilateral Settlement (for interchange fees)
- d. admi.004.001.002 – System Event Notification
- e. camt.054.001.013 – Credit Notification
- f. camt.052.001.08- Balance report

4. Messages supported by the Proxy Module, part of ISO20022 standard:

- a. acmt.022.001.004 – Identification Modification Advice
- b. acmt.023.001.004 – Identification Verification Request
- c. acmt.024.001.003 – Identification Verification Report
- d. pacs.002.001.013 – Status Report

3.3.2. Transaction Processing

This section describes the transaction processing in IPS and the mechanism for updating technical accounts following this process.

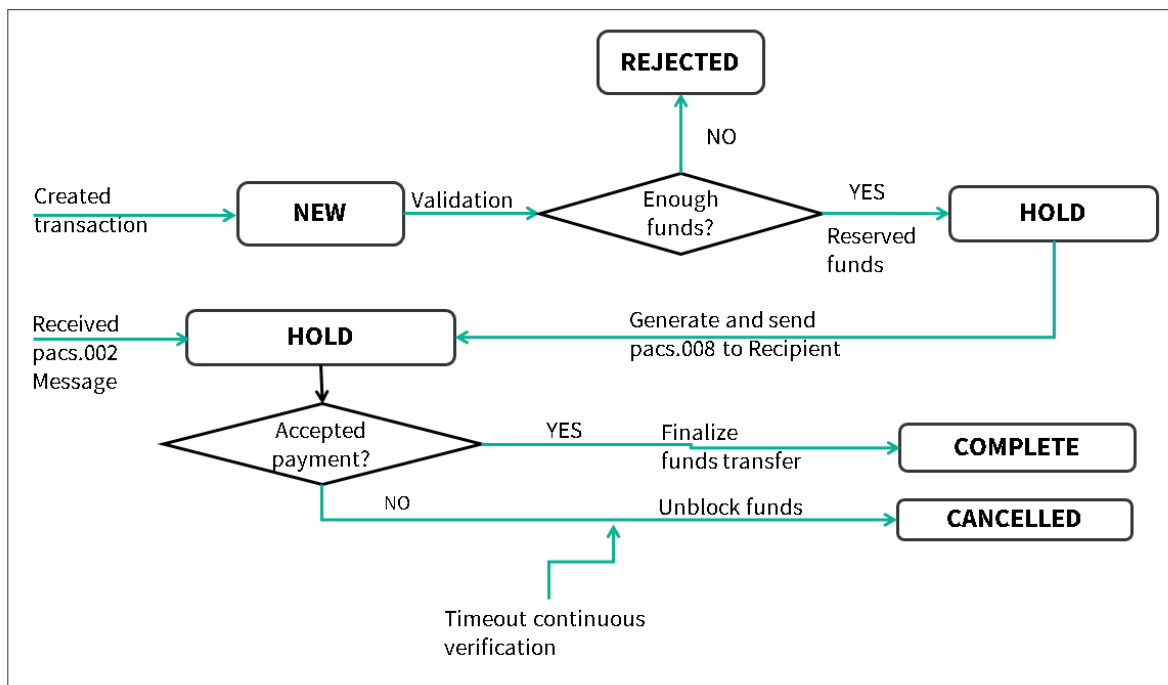


FIGURE 75. TRANSITION OF THE STATUS OF A TRANSACTION IN THE IPS

From the point of view of updating the technical accounts, the processing of transactions in IPS consists of two distinct stages:

1. Reservation of funds for the debtor Participant.
2. Complete successful transaction or cancel it.
 - a. In case of success, the transaction is marked as COMPLETE. It is irreversible and the funds received by the creditor Participant become available for other payments initiated by it (the available balance is increased).
 - b. In case of cancellation of the transaction, the reserved funds are unblocked for the debtor Participant.

To manage this process, IPS uses two distinct entry types in the database, COMPLETE and HOLD, , hereafter referred to as IPS **Conditions**, which can be seen only in the database. Thus, step 1 of the processing means the transaction recording in the HOLD condition. Step 2 consists of transferring the HOLD condition to COMPLETE for success or canceling the HOLD condition to cancel the transaction. This is related to how the system stores the information at a lower level.

In the business sense there is only one account for each IPS participant, however in the database we must maintain a separate entry for HOLD amounts, the reserved funds from step 1, (which can be later moved to COMPLETE or cancelled).

A brief example of successful processing for a transaction from A to B with the amount of 20 is presented below.

	PARTICIPANT A		PARTICIPANT B	
	HOLD	COMPLETE	HOLD	COMPLETE
Initial status	0	250	0	120
Step 1	-20	250	20	120
Step 2	0	230	0	140

Note: Each account and condition is maintained by the IPS as a set of two distinct values: the debited amount and the amount credited. Only the net amount that is dynamically calculated by summing the two amounts when required by the IPS is shown in the table above.

In other words, the net position is a function, the value of which is valued each time it is requested when: verifying the available in transaction processing, displaying in monitoring screens, checking the decrease of the collateral ceiling for indirect participants as well as in other validation and verification cases.

With these details, the available balance for a Participant is not maintained as an individual value, but it is also calculated dynamically whenever required by the formula:

Available Balance (AB) = Net Balance COMPLETE (Credit - Debit) – HOLD Debit Balance.

Thus, considering the above example, after step 1, AB for Participant A becomes 230. AB for participant B is increased by 20 only after completing step 2, because that amount must reach the COMPLETE condition to be considered in the AB calculation.

If the transaction is canceled after step 2, the scenario is as follows.

	PARTICIPANT A		PARTICIPANT B	
	HOLD	COMPLETE	HOLD	COMPLETE
Initial status	0	250	0	120
Step 1	-20	250	20	120
Step 2	0	250	0	120

Practically, records in HOLD condition are removed, and AB for Participant A returns to the initial value (assuming there are no other intercalated transactions).

After sending a transaction, the sender participant will see the updated available balance in the technical account monitoring screen in MMC. The transaction is visible to the receiving participant while in status HOLD as well, both for listing and as an update to the pending credit amount, but the amount is not yet added to the available balance, and it cannot be used until the transaction is completed.

All these account balances are visible in MMC under the Monitoring – Account Positions screen, with the following columns:

Account – the technical account in IPS identifier consisting of participant BIC and currency code

Current balance – the value of the COMPLETE condition described above

Available balance – Net Balance COMPLETE (Credit - Debit) – HOLD Debit Balance; this is the balance that the participant is allowed to use; the participant is not allowed to use the HOLD credit condition value

Debit – debit amount for the current business day for the COMPLETE condition

Credit – credit amount for the current business day for the COMPLETE condition

Pending debit – debit amount currently for HOLD condition

Pending credit – credit amount currently for HOLD condition

Message and Transactions Details

The IPS extracts from the content of the message and saves in the database a set of information related to messages and transactions. They are displayed on the user interface, and some of them can be used to filter transactions in **View** screens. A list of transaction/message information will contain at least:

- Transaction type (e.g., Credit)
- Transaction Channel
- Message typeTran Sequence – (The processing sequence) – The IPS assigns to each processed transaction a uniquely sequenced sequence at the time of step 1 processing. It does not apply to invalid payment messages in the validation phase or those that do not

generate account updates. The uniquely sequenced sequence is used to generate reconciliation messages and to identify transactions processed in a clearing session.

- Transaction reference – retrieved from source message (e.g., TxID field from pacs.008)
- Transaction/message status
 - Complete, Cancelled, Hold, Rejected
 - Additional statuses for recalls: Paid, Paid partial, Refused
 - Additional statuses for RTP: Rejected, Sent, Paid, Paid PART, , Expired, Refused, Accepted, Cancelled.
- The amount of the transaction/message
- The currently paid amount (in case of RTP or recall)
- Currency
- Business DateOrigination Time– in the AcceptanceDateTime field
- Initiation Time - Timestamp when the direct clearing participant receives technically receives the order from client or indirect clearing participant – in the FIToFICstmrCdtTrf - GrpHdr-CreDtTm field
- Merchant Category Code (MCC)
- RTP reference – present inside the EndToEndId tag with the format “RTP-ref”
- The debtor Participant
- Debtor information: name and IBAN account
- The credited Participant
- Creditor information: name and IBAN account
- Processing time (date, time, min, seconds and microseconds):
 - Receive Time - The moment of receiving the message from the sender.
 - Forward Time - The moment of sending the message to the recipient.
 - Reply Receive Time - The moment of receiving the reply from the recipient.
 - Process completion time – which also includes sending the final response to the sender for pacs.008 and pacs.004.
- The content of all the payment messages exchanged with the Participants in the original format (XML).
- Error code – for messages or transactions rejected or cancelled.

The list of information displayed in the MMC for messages contains at least the following fields:

- Internal identifier (Database Registration ID)
- Input/Output sequence
- Type of message
- Direction (Outgoing/Incoming)
- Reference
- Receiving time for received messages or generation time for sent messages
- Sender/ Instructing Agent (BIC)
- Receiver/ Instructed Agent (BIC)
- Status
- Message content in original format (XML).

- Error code

List Transactions and Messages

IPS provides a search function for viewing transactions that allows for filtering transactions by the following criteria:

- Message type
- Transaction type
- Transaction Channel
- Transaction status
- Tran Sequence
- Error Code
- Reference (with flag for wildcards)
- Business Date (between/exactly/starting from/until)
- Origination Time
- Receive Time
- Forward Time
- Reply Receive Time
- Process Completion Time
- Initiation Time
- Merchant Category Code
- RTP Reference
- Currency (fixed 'GEL')
- Processing sequence interval
- Reception interval time (date and time)
- Transaction amount range (between/exactly/less than/greater than)
- Sender
- Receiver
- Creditor Participant
- Debtor Participant
- Participant (checks if selected participant is either sender, receiver, creditor or debtor)
- Debtor name
- Debtor client account (IBAN)
- Creditor name

Creditor client account (IBAN) Message filtering can be done according to the following criteria:

- Reference
- Message type
- Direction
- Sender BIC

- Receiver BIC
- Message Status
- Time (From/To, date and time) - Receiving time for received messages or generation time for sent messages
- Error code

The types of messages that are available in the graphical interface for display are:

1. SCTInst schema messages: pacs.008, pacs.009, pacs.004, pacs.002, camt.056, camt.029, pacs.028, pain.013, pain.014, camt.053.
2. Supplementary messages: Participant List Table – participants.001.
3. Interface messages exchanged with the settlement system.

All of these messages displayed in the graphical user interface can be downloaded by users on the local station in the original format (XML). All messages processed by the system are recorded in a business consistent mode (e.g. grouping together the 5 messages of an ICT - [input:output:receiver reply:receiver confirmation:sender reply], the 3 messages of non-value flow - [input:output:senders reply], the 2 messages of a sweep execution - [request:reply] etc). Status is presented for the business records as well as for the messages. When listing a recall transaction then all associated payments will be displayed as well. Similarly, for a Request To Pay, all transactions will be displayed. The system tracks statuses of recalls and RTPs for reporting purposes in MMC. The matching with transactions and replies is done asynchronously.

3.3.3. Error Codes

After processing the messages and transactions, the IPS assigns to each message an internal error code, as shown in the table below. These error codes are displayed in the graphical interface for operator investigations.

When generating pacs.002 response messages, the error code generated by IPS is mapped to a schema error code.

INTERNAL CODE	PACS.002 MAPPING	DESCRIPTION
000		No error.
100	MS03	Internal processing error.
500	AB05	Timeout error.
501	MS02	Payment declined by the beneficiary Participant.
502	AB08	The recipient Participant is disconnected.
1000	MS03	Generic error.
1001	MS03	Generic validation error.
1002	FF01	The currency is invalid.

INTERNAL CODE	PACS.002 MAPPING	DESCRIPTION
1003	FF01	Amount is invalid (less than or equal to zero).
1004	DNOR	The value in the DbtrAgt field is invalid: - No ACTIVE Participant with this BIC found. - The value is different from InstgAgt. - The value is different from the ordering detected Participant at channel transmission level.
1005	CNOR	The value in the CdtrAgt field is invalid: - No ACTIVE Participant with this BIC found.
1006	RC01	The value in the InstgAgt field is invalid: - No ACTIVE Participant with this BIC found. - The value is different from the ordering detected Participant at channel transmission level.
1007	RC01	Unused.
1008	AC01	The IBAN code of the debtor is invalid.
1009	AC01	The creditor IBAN code is invalid.
1010	AM05	A payment with the same reference is currently processed.
1011	AM05	A payment with the same reference was already processed.
1012	FF01	The settlement date is invalid (IntrBkSttlmDt).
1013	FF01	Invalid XML format.
1014		Unused.
1015	TM01	Invalid Time – Value in the Acceptance DateTime field is older than the current time with the value of the timeout parameter configured in the payment schema.
1016	AG09	Invalid original reference – No transaction with the original reference was found when processing a pacs.002 message.
1017	AG09	Invalid status of the original transaction – When processing a pacs.002 message, the referenced transaction is not in the proper status (message: WAIT_RECEIVER, transaction: HOLD).
1018	RC01	Invalid sender – Sender Participant is not ACTIVE.
1019	RR04	The payment schema is not defined for the currency specified by the message.
1020	AM02	The amount is too high compared to the parameter defined in the payment schema.

INTERNAL CODE	PACS.002 MAPPING	DESCRIPTION
1021	AG10	Sender is blocked (temporarily disabled).
1022	RR04	Payment schema not mapped (either for debtor or creditor)
1026	FF01	Invalid XML field (groups various XML validations not part of default xsd, i.e. specific codes like SvcLvl, Ctgypurp)
1027	RC01	Invalid receiver (receiver is not the system BIC for messages sent by participants)
1028	FF01	Invalid original settlement date (recalls that point to an original date from the future)
1029	RC01	Invalid on us (intra-bank payments sent to IPS)
1030	RC01	Invalid originator (originator BIC missing or invalid)
1033	AB05	Past requested execution day (for RTP flow)
1040	AB05	Missing requested execution day (from RTP message)
1041	MS03	RTP flow is not enabled in payment schema
1042	AB05	Requested execution date is after the max allowed in payment schema
1043	MS03	PmtInflid from RTP is invalid
1044	MS03	Invalid date in PmtInflid from RTP (the id must start with the current date)
1045	MS03	PmtTpInf field present in both group and item description
2000	AM23	Insufficient funds to clear the transaction.
3000	FF01	The Participant does not have any active certificate of DS type for validating messages.
3001	FF01	The XML message does not contain the digital signature in the specified format.
3002	FF01	Digital signature does not protect the entire XML message.
3003	FF01	Digital signature is invalid.
3004	FF01	The certificate used for signing has expired or has been revoked.

For Participants that require it, the schema for pacs.002 validation has been amended to enable the ISO standard [AddtlInf](#). Additional error information field which will contain a description of the error up to 105 characters. The full detail will be also available in the message list for investigations.

3.3.1. External Codes

Category Purpose Codes

The **Category Purpose code (PmtTpInf – CtgyPurp – Cd)** of the SCT Inst Instruction offers information on the high-level nature of the SCT Inst Instruction. All codes part of the ISO standard are accepted

ISO CODE	NAME. DESCRIPTION
BONU	Bonuspayment. Transaction is the payment of a bonus.
CASH	CashManagementTransfer. Transaction is a general cash management instruction.
CBLK	CardBulkClearing. A service that is settling money for a bulk of card transactions, while referring to a specific transaction file or other information like terminal ID, card acceptor ID or other transaction details.
CCRD	CreditCardPayment. Transaction is related to a payment of credit card.
CORT	TradeSettlementPayment. Transaction is related to settlement of a trade, e.g. a foreign exchange deal or a securities transaction.
DCRD	DebitCardPayment. Transaction is related to a payment of debit card.
DIVI	Dividend. Transaction is the payment of dividends.
DVPM	DeliverAgainstPayment. Code used to pre-advise the account servicer of a forthcoming deliver against payment instruction.
EPAY	Epagement. Transaction is related to ePayment.
FCIN	FeeCollectionAndInterest. Transaction is related to the payment of a fee and interest.
FCOL	FeeCollection. A service that is settling card transaction related fees between two parties.
GP2P	PersontoPersonPayment. General Person-to-Person Payment. Debtor and Creditor are natural persons.
GOVT	GovernmentPayment. Transaction is a payment to or from a government department.
HEDG	Hedging. Transaction is related to the payment of a hedging operation.
ICCP	IrrevocableCreditCardPayment. Transaction is reimbursement of credit card payment.
IDCP	IrrevocableDebitCardPayment. Transaction is reimbursement of debit card payment.
INTC	IntraCompanyPayment. Transaction is an intra-company payment, i.e., a payment between two companies belonging to the same group.
INTE	Interest. Transaction is the payment of interest.
LBOX	LockboxTransactions. Transaction is related to identify cash handling via Night Safe or Lockbox by bank or vendor on behalf of a physical store.
LOAN	Loan. Transaction is related to the transfer of a loan to a borrower.
MP2B	Commercial. Mobile P2B Payment

MP2P	Consumer. Mobile P2P
OTHR	OtherPayment. Other payment purpose.
PENS	PensionPayment. Transaction is the payment of pension.
RPRE	Represented. Collection used to re-present previously reversed or returned direct debit transactions.
RRCT	ReimbursementReceivedCreditTransfer. Transaction is related to a reimbursement for commercial reasons of a correctly received credit transfer.
RVPM	ReceiveAgainstPayment. Code used to pre-advise the account servicer of a forthcoming receive against payment instruction.
SALA	SalaryPayment. Transaction is the payment of salaries.
SECU	Securities. Transaction is the payment of securities.
SSBE	SocialSecurityBenefit. Transaction is a social security benefit, ie payment made by a government to support individuals.
SUPP	SupplierPayment. Transaction is related to a payment to a supplier.
TAXS	TaxPayment. Transaction is the payment of taxes.
TRAD	Trade. Transaction is related to the payment of a trade finance transaction.
TREA	TreasuryPayment. Transaction is related to treasury operations. E.g. financial contract settlement.
VATX	ValueAddedTaxPayment. Transaction is the payment of value added tax.
WHLD	WithHolding. Transaction is the payment of withholding tax.
SWEP	CashManagementSweepAccount. Transaction relates to a cash management instruction, requesting a sweep of the account of the Debtor above an agreed floor amount, up to a target or zero balance. The purpose is to move the funds from multiple accounts to a single bank account. Funds can move domestically or across border and more than one bank can be used.
TOPG	CashManagementTopAccount. Transaction relates to a cash management instruction, requesting to top the account of the Creditor above a certain floor amount, up to a target or zero balance. The floor amount, if not pre-agreed by the parties involved, may be specified.
ZABA	CashManagementZeroBalanceAccount. Transaction relates to a cash management instruction, requesting to zero balance the account of the Debtor. Zero Balance Accounts empty or fill the balances in accounts at the same bank, in the same country into or out of a main account each day.
VOST	CrossborderMIPayments. Transaction to be processed as a domestic payment instruction originated from a foreign bank.
FCDT	ForeignCurrencyDomesticTransfer. Foreign Currency Transaction that is processed between two domestic financial institutions.
CIPC	CashInPreCredit. Transaction is a direct debit for a cash order of notes and/or coins.
CONC	CashOutNotesCoins. Transaction is a direct debit for a cash order of notes and/or coins.

CGWV	CarrierGuardedWholesaleValuables. Transaction is a payment towards a Party for the collection of cash by the Cash in Transit company.
SAVG	Savings Transfer to / from savings or to retirement account. Used to identify transfers performed between savings accounts and current accounts held at different PSPs by the same holder, in order to check that the holder of both accounts is the same.
CTDF	DoddFrank1073Payment. Cross border transaction initiated by US natural person that is subject to compliance with Dodd Frank 1073. Category Purpose code for US Consumer Cross Border “Dodd Frank” payments. Used in pacs.008 and pacs.009 messages.

Purpose Code

The **Purpose Code** (Purp – Cd) of the SCT Inst Instruction offers information on the nature of the transaction. All codes part of the ISO standard are accepted and foresees among others a code to flag that an SCT Inst Instruction results from a Request-To-Pay message.

ISO CODE	NAME. DESCRIPTION
BKDF	BankLoanDelayedDrawFunding. Delayed draw funding. Certain issuers may utilize delayed draw loans whereby the lender is committed to fund cash within a specified period once a call is made by the issuer. The lender receives a fee for entering into such a commitment.
BKFE	BankLoanFees. Bank loan fees. Cash activity related to specific bank loan fees, including (a) agent / assignment fees; (b) amendment fees; (c) commitment fees; (d) consent fees; (e) cost of carry fees; (f) delayed compensation fees; (g) facility fees; (h) fronting fees; (i) funding fees; (j) letter of credit assignment fees.
BKFM	BankLoanFundingMemo. Bank loan funding memo. Net cash movement for the loan contract final notification when sent separately from the loan contract final notification instruction.
BKIP	BankLoanAccruedInterestPayment. Accrued interest payments. Specific to bank loans.
BKPP	BankLoanPrincipalPaydown. Principal paydowns. Specific to bank loans
CBLK	CardBulkClearing. A Service that is settling money for a bulk of card transactions, while referring to a specific transaction file or other information like terminal ID, card acceptor ID or other transaction details.
CDCB	CardPaymentWithCashBack. Purchase of Goods and Services with additional Cash disbursement at the POI (Cashback)
CDCD	CashDisbursementCashSettlement. ATM Cash Withdrawal in an unattended or Cash Advance in an attended environment (POI or bank counter).
CDCS	CashDisbursementWithSurcharging. ATM Cash Withdrawal in an unattended or Cash Advance in an attended environment (POI or bank counter) with surcharging.

CDDP	CardDeferredPayment. A combined service which enables the card acceptor to perform an authorisation for a temporary amount and a completion for the final amount within a limited time frame. Deferred Payment is only available in the unattended environment. Examples where this service is widely used are unattended petrol pumps and phone booths.
CDOC	OriginalCredit. A service which allows the card acceptor to effect a credit to a cardholder' account. Unlike a Merchant Refund, an Original Credit is not preceded by a card payment. This service is used for example for crediting winnings from gaming.
CDQC	QuasiCash. Purchase of Goods which are equivalent to cash like coupons in casinos.
ETUP	EPurseTopUp. Transaction is related to a Service that is first reserving money from a card account and then is loading an e-purse application by this amount.
FCOL	FeeCollection. A Service that is settling card transaction related fees between two parties.
MTUP	MobileTopUp. A Service that is first reserving money from a card account and then is loading a prepaid mobile phone amount by this amount.
ACCT	AccountManagement. Transaction moves funds between 2 accounts of same account holder at the same bank.
CASH	CashManagementTransfer. Transaction is a general cash management instruction.
COLL	CollectionPayment. Transaction is a collection of funds initiated via a credit transfer or direct debit.
CSDB	CashDisbursementCashManagement. Transaction is related to cash disbursement.
DEPT	Deposit. Transaction is related to a payment of deposit.
INTC	IntraCompanyPayment.
INTP	IntraPartyPayment. Transaction is a payment between two accounts belonging to the same party (intra-party payment), where party is a natural person (identified by a private ID, not organisation ID).
LIMA	LiquidityManagement. Bank initiated account transfer to support zero target balance management, pooling or sweeping.
NETT	Netting. Transaction is related to a netting operation.
BFWD	BondForward. Cash collateral related to any securities traded out beyond 3 days which include treasury notes, JGBs and Gilts.
CCIR	CrossCurrencyIRS. Cash Collateral related to a Cross Currency Interest Rate Swap, indicating the exchange of fixed interest payments in one currency for those in another.
CCPC	CCPClearedInitialMargin. Cash Collateral associated with an ISDA or Central Clearing Agreement that is covering the initial margin requirements for OTC trades clearing through a CCP.
CCPM	CCPClearedVariationMargin. Cash Collateral associated with an ISDA or Central Clearing Agreement that is covering the variation margin requirements for OTC trades clearing through a CCP.

CCSM	CCPClearedInitialMarginSegregatedCash. CCP Segregated initial margin: Initial margin on OTC Derivatives cleared through a CCP that requires segregation
CRDS	CreditDefaultSwap. Cash collateral related to trading of credit default swap.
CRPR	CrossProduct. Cash collateral related to a combination of various types of trades.
CRSP	CreditSupport. Cash collateral related to cash lending/borrowing; letter of Credit; signing of master agreement.
CRTL	CreditLine. Cash collateral related to opening of a credit line before trading.
EQPT	EquityOption. Cash collateral related to trading of equity option (Also known as stock options).
EQUS	EquitySwap. Cash collateral related to equity swap trades where the return of an equity is exchanged for either a fixed or a floating rate of interest.
EXPT	ExoticOption. Cash collateral related to trading of an exotic option for example a non-standard option.
EXTD	ExchangeTradedDerivatives. Cash collateral related to trading of exchanged traded derivatives in general (Opposite to Over the Counter (OTC)).
FIXI	FixedIncome. Cash collateral related to a fixed income instrument
FWBC	ForwardBrokerOwnedCashCollateral. Cash collateral payment against a Master Forward Agreement (MFA) where the cash is held in a segregated account and is not available for use by the client. Includes any instruments with a forward settling date such TBAs, repurchase agreements and bond forwards
FWCC	ForwardClientOwnedCashCollateral. Cash collateral payment against a Master Forward Agreement (MFA) where the cash is owned and may be used by the client when returned. Includes any instruments with a forward settling date such TBAs, repurchase agreements and bond forwards
FWSB	ForwardBrokerOwnedCashCollateralSegregated. Any cash payment related to the collateral for a Master Agreement forward, which is segregated, and not available for use by the client. Example master agreement forwards include TBA, repo and Bond Forwards.
FWSC	ForwardClientOwnedSegregatedCashCollateral. Any cash payment related to the collateral for a Master agreement forward, which is owned by the client and is available for use by the client when it is returned to them from the segregated account. Example master agreement forwards include TBA, repo and Bond Forwards.
MARG	DailyMarginOnListedDerivatives
MBSB	MBSBrokerOwnedCashCollateral. MBS Broker Owned Segregated (40Act/Dodd Frank) Cash Collateral - Any cash payment related to the collateral for a Mortgage Back Security, which is segregated, and not available for use by the client.
MBSC	MBSCClientOwnedCashCollateral. MBS Client Owned Cash Segregated (40Act/Dodd Frank) Cash Collateral - Any cash payment related to the collateral for a Mortgage Back Security, which is owned by the client and is available for use by the client when it is returned to them from the segregated account.

MGCC	FuturesInitialMargin. Initial futures margin. Where such payment is owned by the client and is available for use by them on return.
MGSC	FuturesInitialMarginClientOwnedSegregatedCashCollateral. Margin Client Owned Segregated Cash Collateral - Any cash payment related to the collateral for initial futures margin, which is owned by the client and is available for use by the client when it is returned to them from the segregated account.
OCCC	ClientOwnedOCCPledgedCollateral. Client owned collateral identified as eligible for OCC pledging.
OPBC	OTCOptionBrokerOwnedCashCollateral. Cash collateral payment for OTC options associated with an FCM agreement. Where such payment is segregated and not available for use by the client.
OPCC	OTCOptionClientOwnedCashCollateral. Cash collateral payment for OTC options associated with an FCM agreement. Where such payment is not segregated and is available for use by the client upon return.
OPSB	OTCOptionBrokerOwnedSegregatedCashCollateral. Option Broker Owned Segregated Cash Collateral - Any cash payment related to the collateral for an OTC option, which is segregated, and not available for use by the client.
OPSC	OTCOptionClientOwnedCashSegregatedCashCollateral. Option Client Owned Cash Segregated Cash Collateral - Any cash payment related to the collateral for an OTC option, which is owned by the client and is available for use by the client when it is returned to them from the segregated account.
OPTN	FXOption. Cash collateral related to trading of option on Foreign Exchange.
OTCD	OTCDerivatives. Cash collateral related to Over-the-counter (OTC) Derivatives in general for example contracts which are traded and privately negotiated.
REPO	RepurchaseAgreement. Cash collateral related to a repurchase agreement transaction.
RPBC	BilateralRepoBrokerOwnedCollateral. Bi-lateral repo broker owned collateral associated with a repo master agreement – GMRA or MRA Master Repo Agreements.
RPCC	RepoClientOwnedCollateral. Repo client owned collateral associated with a repo master agreement – GMRA or MRA Master Repo Agreements.
RPSB	BilateralRepoBrokerOwnedSegregatedCashCollateral. Bi-lateral repo broker owned segregated cash collateral associated with a repo master agreement.
RPSC	BilateralRepoClientOwnedSegregatedCashCollateral. Repo client owned segregated collateral associated with a repo master agreement.
RVPO	ReverseRepurchaseAgreement. Cash collateral related to a reverse repurchase agreement transaction.
SBSC	SecuritiesBuySellSellBuyBack. Cash collateral related to a Securities Buy Sell Sell Buy Back.
SCIE	SingleCurrencyIRSExotic. Cash collateral related to Exotic single currency interest rate swap.
SCIR	SingleCurrencyIRS. Cash collateral related to Single Currency Interest Rate Swap.

SCRCP	SecuritiesCrossProducts. Cash collateral related to Combination of securities-related exposure types.
SHBC	BrokerOwnedCollateralShortSale. Short Sale broker owned collateral associated with a prime broker agreement.
SHCC	ClientOwnedCollateralShortSale. Short Sale client owned collateral associated with a prime brokerage agreement
SHSL	ShortSell. Cash Collateral related to a Short Sell
SLEB	SecuritiesLendingAndBorrowing. Cash collateral related to Securities lending and borrowing.
SLOA	SecuredLoan. Cash collateral related to a Secured loan.
SWBC	SwapBrokerOwnedCashCollateral. Cash collateral payment for swaps associated with an ISDA agreement. . Where such payment is segregated and not available for use by the client. Includes any cash collateral payments made under the terms of a CSA agreement for instruments such as swaps and FX forwards.
SWCC	SwapClientOwnedCashCollateral. Cash collateral payment for swaps associated with an ISDA agreement. Where such payment is not segregated and is available for use by the client upon return. Includes any cash collateral payments made under the terms of a CSA agreement for instruments such as swaps and FX forwards.
SWPT	Swaption. Cash collateral related to an option on interest rate swap.
SWSB	SwapsBrokerOwnedSegregatedCashCollateral. Swaps Broker Owned Segregated Cash Collateral - Any cash payment related to the collateral for Swap margin, which is segregated, and not available for use by the client. This includes any collateral identified in a CSA agreement such as Swap or FX Forward collateral.
SWSC	SwapsClientOwnedSegregatedCashCollateral. Swaps Client Owned Segregated Cash Collateral - Any cash payment related to the collateral for Swap margin, which is owned by the client and is available for use by the client when returned from the segregated account. This includes any collateral identified in a CSA agreement such as Swap or FX Forward collateral.
TBAS	ToBeAnnounced. Cash collateral related to a To Be Announced (TBA).
TBBC	TBABrokerOwnedCashCollateral. Cash collateral payment (segregated) for TBA securities associated with a TBA Master Agreement. Where such payment is segregated and not available for use by the client.
TBCC	TBAClientOwnedCashCollateral. Cash collateral payment (for use by client)for TBA securities associated with a TBA Master Agreement. Where such payment is not segregated and is available for use by the client upon return.
TRCP	TreasuryCrossProduct. Cash collateral related to a combination of treasury-related exposure types.
AGRT	AgriculturalTransfer. Transaction is related to the agricultural domain.
AREN	AccountsReceivablesEntry. Transaction is related to a payment associated with an Account Receivable Entry
BEXP	BusinessExpenses. Transaction is related to a payment of business expenses.

BOCE	BackOfficeConversionEntry. Transaction is related to a payment associated with a Back Office Conversion Entry
COMC	CommercialPayment. Transaction is related to a payment of commercial credit or debit. (formerly CommercialCredit)
CPYR	Copyright. Transaction is payment of copyright.
GDDS	PurchaseSaleOfGoods. Transaction is related to purchase and sale of goods.
GDSV	PurchaseSaleOfGoodsAndServices. Transaction is related to purchase and sale of goods and services.
GSCB	PurchaseSaleOfGoodsAndServicesWithCashBack. Transaction is related to purchase and sale of goods and services with cash back.
LICF	LicenseFee. Transaction is payment of a license fee.
MP2B	MobileP2BPayment. A service which enables a user to use an app on its mobile to pay a merchant or other business payees by initiating a payment message. Within this context, the account information or an alias of the payee might be transported through different channels to the app, for example QR Code, NFC, Bluetooth, other Networks.
POPE	PointOfPurchaseEntry. Transaction is related to a payment associated with a Point of Purchase Entry.
ROYA	Royalties. Transaction is the payment of royalties.
SCVE	PurchaseSaleOfServices. Transaction is related to purchase and sale of services.
SERV	ServiceCharges. Transaction is related to service charges charged by a service provider.
SUBS	Subscription. Transaction is related to a payment of information or entertainment services either in printed or electronic form.
SUPP	SupplierPayment. Transaction is related to a payment to a supplier.
TRAD	Commercial. Transaction is related to a trade services operation.
CHAR	CharityPayment. Transaction is a payment for charity reasons.
COMT	ConsumerThirdPartyConsolidatedPayment. Transaction is a payment used by a third party who can collect funds to pay on behalf of consumers, i.e., credit counseling or bill payment companies.
MP2P	MobileP2PPayment. A service which enables a user to use an app on its mobile to initiate moving funds from his/her bank account to another person's bank account while not using the account number but an alias information like an MSISDN as account addressing information in his/her app.
ECPG	GuaranteedEPayment. E-Commerce payment with payment guarantee of the issuing bank.
ECPR	EPaymentReturn. E-Commerce payment return.
ECPU	NonGuaranteedEPayment. E-Commerce payment without payment guarantee of the issuing bank.
EPAY	Epayment. Transaction is related to ePayment.

CLPR	CarLoanPrincipalRepayment. Transaction is a payment of car loan principal payment.
COMP	CompensationPayment. Transaction is related to the payment of a compensation relating to interest loss/value date adjustment and can include fees.
DBTC	DebitCollectionPayment. Collection of funds initiated via a debit transfer.
GOVI	GovernmentInsurance. Transaction is related to a payment of government insurance.
HLRP	PropertyLoanRepayment. Transaction is related to a payment of property loan.
HLST	PropertyLoanSettlement. Transaction is related to the settlement of a property loan.
INPC	InsurancePremiumCar. Transaction is a payment of car insurance premium.
INPR	InsurancePremiumRefund. Transaction is related to an insurance premium refund.
INSC	PaymentOfInsuranceClaim. Transaction is related to the payment of an insurance claim.
INSU	InsurancePremium. Transaction is payment of an insurance premium.
INTE	Interest. Transaction is payment of interest.
LBRI	LaborInsurance.
LIFI	LifeInsurance. Transaction is a payment of life insurance.
LOAN	Loan. Transaction is related to transfer of loan to borrower.
LOAR	LoanRepayment. Transaction is related to repayment of loan to lender.
PENO	PaymentBasedOnEnforcementOrder. Payment based on enforcement orders except those arising from judicial alimony decrees.
PPTI	PropertyInsurance. Transaction is a payment of property insurance.
RELG	RentalLeaseGeneral. Transaction is for general rental/lease.
RINP	RecurringInstallmentPayment. Transaction is related to a payment of a recurring installment made at regular intervals.
TRFD	TrustFund. Transaction is related to a payment of a trust fund.
FORW	ForwardForeignExchange. FX trades with a value date in the future.
FXNT	ForeignExchangeRelatedNetting. FX netting if cash is moved by separate wire instead of within the closing FX instruction
ADMG	AdministrativeManagement. Transaction is related to a payment associated with administrative management.
ADVA	AdvancePayment. Transaction is an advance payment.
BCDM	BearerChequeDomestic. Transaction is the payment of a domestic bearer cheque.
BCFG	BearerChequeForeign. Transaction is the payment of a foreign bearer cheque.
BLDM	BuildingMaintenance. Transaction is related to a payment associated with building maintenance.
BNET	BondForwardNetting. Bond Forward pair-off cash net movement

CBFF	CapitalBuilding. Transaction is related to capital building fringe fortune, ie capital building in general
CBFR	CapitalBuildingRetirement. Transaction is related to capital building fringe fortune for retirement
CCRD	CreditCardPayment. Transaction is related to a payment of credit card account.
CDBL	CreditCardBill. Transaction is related to a payment of credit card bill.
CFEE	CancellationFee. Transaction is related to a payment of cancellation fee.
CGDD	CardGeneratedDirectDebit. Transaction is related to a direct debit where the mandate was generated by using data from a payment card at the point of sale.
CORT	TradeSettlementPayment. Transaction is related to settlement of a trade, e.g. a foreign exchange deal or a securities transaction.
COST	Costs. Transaction is related to payment of costs.
CPKC	CarparkCharges. Transaction is related to carpark charges.
DCRD	DebitCardPayment. Transaction is related to a debit card payment.
DSMT	PrintedOrderDisbursement. Transaction is the payment of a disbursement due to a specific type of printed order for a payment of a specified sum, issued by a bank or a post office (Zahlungsanweisung zur Verrechnung)
DVPM	DeliverAgainstPayment. Code used to pre-advise the account servicer of a forthcoming deliver against payment instruction.
EDUC	Education. Transaction is related to a payment of study/tuition fees.
FACT	FactorUpdateRelatedPayment. Payment related to a factor update
FAND	FinancialAidInCaseOfNaturalDisaster. Financial aid by State authorities for abolition of consequences of natural disasters.
FCPM	LatePaymentOfFeesAndCharges. Transaction is the payment for late fees & charges. E.g Credit card charges
FEES	PaymentOfFees. Payment of fees/charges.
GIFT	Gift. Payment with no commercial or statutory purpose.
GOVT	GovernmentPayment. Transaction is a payment to or from a government department.
ICCP	IrrevocableCreditCardPayment. Transaction is reimbursement of credit card payment.
IDCP	IrrevocableDebitCardPayment. Transaction is reimbursement of debit card payment.
IHRP	InstalmentHirePurchaseAgreement. Transaction is payment for an installment/hire-purchase agreement.
INSM	Installment. Transaction is related to a payment of an installment.
IVPT	InvoicePayment. Transaction is the payment for invoices.
MCDM	MultiCurrenyChequeDomestic. Transaction is the payment of a domestic multi-currency cheque

MCFG	MultiCurrenyChequeForeign. Transaction is the payment of a foreign multi-currency cheque
MSVC	MultipleServiceTypes. Transaction is related to a payment for multiple service types.
NOWS	NotOtherwiseSpecified. Transaction is related to a payment for type of services not specified elsewhere.
OCDM	OrderChequeDomestic. Transaction is the payment of a domestic order cheque
OCFG	OrderChequeForeign. Transaction is the payment of a foreign order cheque
OFEE	OpeningFee.Transaction is related to a payment of opening fee.
OTHR	Other. Other payment purpose.
PADD	PreauthorizedDebit. Transaction is related to a pre-authorized debit origination
PTSP	PaymentTerms. Transaction is related to payment terms specifications
RCKE	RepresentedCheckEntry. Transaction is related to a payment associated with a re-presented check entry
RCPT	ReceiptPayment. Transaction is related to a payment of receipt.
REBT	Rebate. Transaction is the payment of a rebate.
REFU	Refund. Transaction is the payment of a refund.
RENT	Rent. Transaction is the payment of rent.
REOD	AccountOverdraftRepayment. Transaction is for account overdraft repayment.
RIMB	ReimbursementOfAPreviousErroneousTransaction. Transaction is related to a reimbursement of a previous erroneous transaction.
RPNT	BilateralRepoInternetNetting. Bi-lateral repo interest net/bulk payment at rollover/pair-off or other closing scenarios where applicable
RRBN	RoundRobin. Cash payment resulting from a Round Robin
RRCT	ReimbursementReceivedCreditTransfer. Transaction is related to a reimbursement for commercial reasons of a correctly received credit transfer.
RRTP	RelatedRequestToPay. Transaction is related to a Request to Pay.
RVPM	ReceiveAgainstPayment. Code used to pre-advise the account servicer of a forthcoming receive against payment instruction.
SLPI	PaymentSlipInstruction. Transaction is payment of a well formatted payment slip.
SPLT	SplitPayments. Split payments. To be used when cash and security movements for a security trade settlement are instructed separately.
STDY	Study.
TBAN	TBAPairOffNetting. TBA pair-off cash wire net movement
TBIL	TelecommunicationsBill. Transaction is related to a payment of telecommunications related bill.
TCSC	TownCouncilServiceCharges. Transaction is related to a payment associated with charges levied by a town council.

TELI	TelephoneInitiatedTransaction. Transaction is related to a payment initiated via telephone.
TMPG	TMPGClaimPayment. Cash payment resulting from a TMPG Claim
TPRI	TriPartyRepoInterest. Tri-Party Repo related interest
TPRP	TriPartyRepoNetting. Tri-party Repo related net gain/loss cash movement
TRNC	TruncatedPaymentSlip. Transaction is payment of a beneficiary prefilled payment slip where beneficiary to payer information is truncated.
TRVC	TravellerCheque. Transaction is the payment of a travellers cheque.
WEBI	InternetInitiatedTransaction. Transaction is related to a payment initiated via internet.
IPAY	InstantPayments. Transaction in which the amount is available to the payee immediately.
IPCA	InstantPaymentsCancellation. Transaction in which the Return of the amount is fully returned.
IPDO	InstantPaymentsForDonations. Transaction in which the amount is available to the payee immediately, done for donations, with sending the address data of the payer.
IPEA	InstantPaymentsInECommerceWithoutAddressData. Transaction in which the amount is available to the payee immediately, done in E-commerce, without sending the address data of the payer.
IPEC	InstantPaymentsInECommerceWithAddressData. Transaction in which the amount is available to the payee immediately, done in E-commerce, with sending the address data of the payer.
IPEW	InstantPaymentsInECommerce. Transaction in which the amount is available to the payee immediately, done in E-commerce.
IPPS	InstantPaymentsAtPOS. Transaction in which the amount is available to the payee immediately, done at POS.
IPRT	InstantPaymentsReturn. Transaction in which the Return of the amount is fully or partial returned.
IPU2	InstantPaymentsUnattendedVendingMachineWith2FA. Transaction is made via an unattended vending machine by using 2-factor-authentication.
IPUW	InstantPaymentsUnattendedVendingMachineWithout2FA. Transaction is made via an unattended vending machine by without using 2-factor-authentication.
ANNI	Annuity. Transaction settles annuity related to credit, insurance, investments, other.
CAFI	CustodianManagementFeeInhouse. Transaction is the payment of custodian account management fee where custodian bank and current account servicing bank coincide
CFDI	CapitalFallingDueInhouse. Transaction is the payment of capital falling due where custodian bank and current account servicing bank coincide
CMDT	CommodityTransfer. Transaction is payment of commodities.
DERI	Derivatives. Transaction is related to a derivatives transaction

DIVD	Dividend. Transaction is payment of dividends.
FREX	ForeignExchange. Transaction is related to a foreign exchange operation.
HEDG	Hedging. Transaction is related to a hedging operation.
INVS	InvestmentAndSecurities. Transaction is for the payment of mutual funds, investment products and shares
PRME	PreciousMetal. Transaction is related to a precious metal operation.
SAVG	Savings. Transfer to / from savings or to retirement account.
SECU	Securities. Transaction is the payment of securities.
SEPI	SecuritiesPurchaseInhouse. Transaction is the payment of a purchase of securities where custodian bank and current account servicing bank coincide
TREA	TreasuryPayment. Transaction is related to treasury operations.
UNIT	UnitTrustPurchase. Transaction is purchase of Unit Trust
FNET	FuturesNettingPayment. Cash associated with a netting of futures payments. Refer to CCPM codeword for netting of initial and variation margin through a CCP.
FUTR	Futures. Cash related to futures trading activity.
ANTS	AnesthesiaServices. Transaction is a payment for anesthesia services.
CVCF	ConvalescentCareFacility. Transaction is a payment for convalescence care facility services.
DMEQ	DurableMedicalequipment. Transaction is a payment is for use of durable medical equipment.
DNTS	DentalServices. Transaction is a payment for dental services.
HLTC	HomeHealthCare. Transaction is a payment for home health care services.
HLTI	HealthInsurance. Transaction is a payment of health insurance.
HSPC	HospitalCare. Transaction is a payment for hospital care services.
ICRF	IntermediateCareFacility. Transaction is a payment for intermediate care facility services.
LTCF	LongTermCareFacility. Transaction is a payment for long-term care facility services.
MAFC	MedicalAidFundContribution. Transaction is contribution to medical aid fund.
MARF	MedicalAidRefund. Transaction is related to a medical aid refund.
MDCS	MedicalServices
VIEW	VisionCare. Transaction is a payment for vision care services.
CDEP	CreditDefaultEventPayment. Payment related to a credit default event
SWFP	SwapContractFinalPayment. Final payments for a swap contract
SWPP	SwapContractPartialPayment. Partial payment for a swap contract
SWRS	SwapContractResetPayment. Reset payment for a swap contract
SWUF	SwapContractUpfrontPayment. Upfront payment for a swap contract

ADCS	AdvisoryDonationCopyrightServices. Payments for donation, sponsorship, advisory, intellectual and other copyright services.
AEMP	ActiveEmploymentPolicy. Payment concerning active employment policy.
ALLW	Allowance. Transaction is the payment of allowances.
ALMY	AlimonyPayment. Transaction is the payment of alimony.
BBSC	BabyBonusScheme. Transaction is related to a payment made as incentive to encourage parents to have more children
BECH	ChildBenefit. Transaction is related to a payment made to assist parent/guardian to maintain child.
BENE	UnemploymentDisabilityBenefit. Transaction is related to a payment to a person who is unemployed/disabled.
BONU	BonusPayment.
CCHD	CashCompensationHelplessnessDisability. Payments made by Government institute related to cash compensation, helplessness, disability. These payments are made by the Government institution as a social benefit in addition to regularly paid salary or pension.
COMM	Commission. Transaction is payment of commission.
CSLP	CompanySocialLoanPaymentToBank. Transaction is a payment by a company to a bank for financing social loans to employees.
GFRP	GuaranteeFundRightsPayment. Compensation to unemployed persons during insolvency procedures.
GVEA	AustrianGovernmentEmployeesCategoryA. Transaction is payment to category A Austrian government employees.
GVEB	AustrianGovernmentEmployeesCategoryB. Transaction is payment to category B Austrian government employees.
GVEC	AustrianGovernmentEmployeesCategoryC. Transaction is payment to category C Austrian government employees.
GVED	AustrianGovernmentEmployeesCategoryD. Transaction is payment to category D Austrian government employees.
GWLT	GovernmentWarLegislationTransfer. Payment to victims of war violence and to disabled soldiers.
HREC	HousingRelatedContribution. Transaction is a contribution by an employer to the housing expenditures (purchase, construction, renovation) of the employees within a tax free fringe benefit system
PAYR	Payroll. Transaction is related to the payment of payroll.
PEFC	PensionFundContribution. Transaction is contribution to pension fund.
PENS	PensionPayment. Transaction is the payment of pension.
PRCP	PricePayment. Transaction is related to a payment of a price.
RHBS	RehabilitationSupport. Benefit for the duration of occupational rehabilitation.

SALA	SalaryPayment. Transaction is the payment of salaries.
SPSP	SalaryPensionSumPayment. Salary or pension payment for more months in one amount or a delayed payment of salaries or pensions.
SSBE	SocialSecurityBenefit. Transaction is a social security benefit, ie payment made by a government to support individuals.
LBIN	LendingBuyInNetting. Net payment related to a buy-in. When an investment manager is bought in on a sell trade that fails due to a failed securities lending recall, the IM may seize the underlying collateral to pay for the buy-in. Any difference between the value of the collateral and the sell proceeds would be paid or received under this code
LCOL	LendingCashCollateralFreeMovement. Free movement of cash collateral. Cash collateral paid by the borrower is done separately from the delivery of the shares at loan opening or return of collateral done separately from return of the loaned security. Note: common when the currency of the security is different the currency of the cash collateral.
LFEE	LendingFees. Fee payments, other than rebates, for securities lending. Includes (a) exclusive fees; (b) transaction fees; (c) custodian fees; (d) minimum balance fees.
LMEQ	LendingEquityMarkedToMarketCashCollateral. Cash collateral payments resulting from the marked-to-market of a portfolio of loaned equity securities
LMFI	LendingFixedIncomeMarkedToMarketCashCollateral. Cash collateral payments resulting from the marked-to-market of a portfolio of loaned fixed income securities.
LMRK	LendingUnspecifiedTypeOfMarkedToMarketCashCollateral. Cash collateral payments resulting from the marked-to-market of a portfolio of loaned securities where the instrument types are not specified.
LREB	LendingRebatePayments. Securities lending rebate payments
LREV	LendingRevenuePayments. Revenue payments made by the lending agent to the client
LSFL	LendingClaimPayment. Payments made by a borrower to a lending agent to satisfy claims made by the investment manager related to sell fails from late loan recall deliveries
ESTX	EstateTax. Transaction is related to a payment of estate tax.
FWLV	ForeignWorkerLevy. Transaction is related to a payment of Foreign Worker Levy
GSTX	GoodsServicesTax. Transaction is the payment of Goods & Services Tax
HSTX	HousingTax. Transaction is related to a payment of housing tax.
INTX	IncomeTax. Transaction is related to a payment of income tax.
NITX	NetIncomeTax. Transaction is related to a payment of net income tax.
PTXP	PropertyTax. Transaction is related to a payment of property tax.
RDTX	RoadTax. Transaction is related to a payment of road tax.
TAXS	TaxPayment. Transaction is the payment of taxes.
VATX	ValueAddedTaxPayment. Transaction is the payment of value added tax.

WHLD	WithHolding. Transaction is related to a payment of withholding tax.
TAXR	TaxRefund. Transaction is the refund of a tax payment or obligation.
B112	TrailerFeePayment. US mutual fund trailer fee (12b-1) payment
BR12	TrailerFeeRebate. US mutual fund trailer fee (12b-1) rebate payment
TLRF	NonUSMutualFundTrailerFeePayment. Any non-US mutual fund trailer fee (retrocession) payment (use ISIN to determine onshore versus offshore designation)
TLRR	NonUSMutualFundTrailerFeeRebatePayment. Any non-US mutual fund trailer fee (retrocession) rebate payment (use ISIN to determine onshore versus offshore designation)
AIRB	Air. Transaction is a payment for air transport related business.
BUSB	Bus. Transaction is a payment for bus transport related business.
FERB	Ferry. Transaction is a payment for ferry related business.
RLWY	Railway. Transaction is a payment for railway transport related business.
TRPT	RoadPricing. Transaction is for the payment to top-up pre-paid card and electronic road pricing for the purpose of transportation
ELEC	ElectricityBill. Transaction is related to a payment of electricity bill.
ENRG	Energies. Transaction is related to a utility operation.
GASB	GasBill. Transaction is related to a payment of gas bill.
NWCH	NetworkCharge. Transaction is related to a payment of network charges.
NWCM	NetworkCommunication. Transaction is related to a payment of network communication.
CBTV	CableTVBill. Transaction is related to a payment of cable TV bill.
OTLC	OtherTelecomRelatedBill. Transaction is related to a payment of other telecom related bill.
PHON	TelephoneBill. Transaction is related to a payment of telephone bill.
UBIL	Utilities. Transaction is for the payment to common utility provider that provide gas, water and/or electricity.
WTER	WaterBill. Transaction is related to a payment of water bill.
BOND	Bonds. Securities Lending-Settlement of Bond transaction.
CABD	CorporateActions-Bonds. Securities Lending-Settlement of Corporate Actions: Bonds transactions.
CAEQ	CorporateActions-Equities. Securities Lending-Settlement of Corporate Actions: Equities transactions.
CBCR	CreditCard. Card Settlement-Settlement of Credit Card transactions.
DBCR	DebitCard. Card Settlement-Settlement of Debit Card transactions.
DICL	Diners. Card Settlement-Settlement of Diners transactions.
EQTS	Equities. Securities Lending-Settlement of Equities transactions.

FLCR	FleetCard. Card Settlement-Settlement of Fleet transactions.
EFTC	LowValueCredit
EFTD	LowValueDebit. Utilities-Settlement of Low value Debit transactions.
MOMA	MoneyMarket. Securities Lending-settlement of Money Market PCH.
RAPI	RapidPaymentInstruction. Instant Payments-Settlement of Rapid Payment Instruction (RPI) transactions.
GAMB	GamblingOrWageringPayment. General-Payments towards a purchase or winnings received from gambling, betting or other wagering activities.
LOTT	LotteryPayment. General-Payment towards a purchase or winnings received from lottery activities.
AMEX	Amex. Card Settlement-Settlement of AMEX transactions.
SASW	ATM. Card Settlement-Settlement of ATM transactions.
AUCO	AuthenticatedCollections. Utilities-Settlement of Authenticated Collections transactions.
PCOM	PropertyCompletionPayment. Final payment to complete the purchase of a property.
PDEP	PropertyDeposit. Payment of the deposit required towards purchase of a property.
PLDS	PropertyLoanDisbursement. Payment of funds from a lender as part of the issuance of a property loan.
PLRF	PropertyLoanRefinancing. Transfer or extension of a property financing arrangement to a new deal or loan provider, without change of ownership of property.
GAFA	GovernmentFamilyAllowance. Salary and Benefits-Allowance from government to support family.
GAHO	GovernmentHousingAllowance. Salary and Benefits-Allowance from government to individuals to support payments of housing.
CPEN	CashPenalties. Cash penalties related to securities transaction, including CSDR Settlement Discipline Regime.
DEPD	DependentSupportPayment. Transaction is related to a payment concerning dependent support, for example child support or support for a person substantially financially dependent on the support provider.
RETL	RetailPayment. Retail payment including e-commerce and online shopping.
DEBT	ChargesBorneByDebtor. Purpose of payment is the settlement of charges payable by the debtor in relation to an underlying customer credit transfer.
RMCO	RegisteredMandate. Utilities-Settlement of Unauthenticated Collections transactions.

Local Instrument

The **Local Instrument** code (PmtTpInf – LclInstrm – Cd) is the mandatory identification Code with value INST that identifies the Instant Credit Transfers from other payments.

ISO CODE	NAME. DESCRIPTION
INST	InstantCreditTransfer. Transaction is related to an Instant Credit Transfer. Use case example: SEPA Instant Credit Transfer (SCT Inst)

Clearing System

The Clearing System code (SttlmInf-ClrSys-Cd) will be created by the Operator after system setup.

ISO CODE	NAME. DESCRIPTION
No default values	

Service Level

The Service Level code (SttlmInf – SvcLvl - Cd) will be created by the Operator after system setup.

ISO CODE	NAME. DESCRIPTION
No default values	

Recall Reasons

The **Recall Reason** codes (camt.056, CxlRsn – Rsn – Cd) explain the reason for the Recall for a SCT Inst transaction. The Originator PSP defines this code and can be used by the Beneficiary PSP to inform the Beneficiary about the reason for the debit of the Payment Account of the Beneficiary.

ISO CODE	DESCRIPTION
DUPL	Duplicate payment. Duplicate sending. (10 Banking Business Days)
TECH	Technical problem. Technical problems resulting in erroneous SEPA instant credit transfer transactions.(10 Banking Business Days)
FRAD	Fraudulent Origin. Fraudulent originated instant credit transfer.(13 months)
AC03	Invalid Creditor Account Number. Wrong unique identifier of the Beneficiary account.
CUST	Requested by Customer. By request of the Originator without any reason specified.
AM09	Wrong amount. Wrong amount

Recall Nack Reason

The **Recall Nack Reason** codes (camt.029, CxlStsRsnInf – Rsn – Cd) define the Reason code for non-acceptance of the Recall. The values are described in the table below.

ISO CODE	DESCRIPTION
AC04	Closed account number. Account closed. Note: This code cannot be used in certain SEPA countries for reasons of data protection. In this case MS03 can be used as alternative. (???)
AM04	Insufficient Funds. Insufficient funds on the account. Note: This code cannot be used in certain SEPA countries for reasons of data protection. CUST should be used instead.
AM09	Wrong amount. Wrong amount
ARDT	The Transaction has Already Been Returned. Already returned transaction.
CUST	Requested by Customer. Refusal by the beneficiary.
LEGL	Legal Decision. Legal reasons.
NOAS	No Answer From Customer. No response from Beneficiary.
NOOR	No Original Transaction Received. Original SEPA Instant Credit Transfer never received.

Return Reasons

The **Return Reason** codes (pacs.004, RtrRsnInf – Rsn – Cd) define the Reason code for acceptance of the Recall. The values are described in the table below.

ISO CODE	DESCRIPTION
AC01	Incorrect Account Number. Account identifier invalid or incorrect (i.e., invalid IBAN or account number does not exist).
FOCR	Following Cancellation Request. Positive answer to the Recall.

Reject Reasons

The **Reject Reasons** code (pacs.002, StsRsnInf – Rsn – Cd) for non-acceptance of a SCT Inst identify the reason for the non-acceptance of the SCT Inst. The values accepted by the SCT Inst scheme are presented in the table below.

ISO CODE	NAME. DESCRIPTION
AB05	Timeout creditor agent. Transaction stopped due to timeout at the creditor agent.
AB06	Timeout Instructed Agent. Transaction stopped due to timeout at the Instructed Agent.

AB07	Offline agent. Agent of message is not online. Generic usage if it cannot be determined which agent is not online.
AB08	Offline creditor agent. Creditor Agent is not online.
AB09	Error Creditor Agent. Transaction stopped due to error at the Creditor Agent.
AB10	Error Instructed Agent. Transaction stopped due to error at the Instructed Agent.
AC01	Incorrect Account Number. Account identifier invalid or incorrect (i.e., invalid IBAN or account number does not exist).
AC04	Closed account number. Account closed.
AC06	Blocked account. Account blocked.
AG01	Transaction Forbidden. Credit transfer forbidden on this account.
AG02	Invalid Bank Operation Code. Operation code / transaction code incorrect or invalid file format.
AG09	Payment Not Received. Original payment never received.
AG10	Agent Suspended. Agent of message is suspended from the Real Time Payment system. This code must be used when it cannot be determined whether it is the Beneficiary PSP itself or another agent in the chain which is (temporarily) suspended.
AG11	Creditor Agent Suspended. Creditor Agent of message is suspended from the Real Time Payment system.
AM02	Not Allowed Amount. Amount exceeds the maximum authorized amount for SCT Inst. [This code will no longer available starting with October 2025 due to regulatory changes of the 2025 SCT Inst Rulebook.]
AM05	Duplication. Duplicate payment.
AM23	Amount Exceeds Settlement Limit. Transaction amount exceeds settlement limit.
BE04	Missing Creditor Address. Account address invalid.
CNOR	Creditor Bank is Not Registered. Beneficiary PSP is not registered under this BIC in the CSM.
DNOR	Debtor Bank is Not Registered. Originator PSP is not registered under this BIC in the CSM.
FF01	Invalid File Format. Operation / transaction code incorrect or invalid file format.
MD07	End Customer Deceased. Beneficiary deceased. Note: This code cannot be used in certain SEPA countries for reasons of data protection. In this case MS03 can be used as alternative.
MS02	Not Specified Reason Customer Generated. By order of the beneficiary.
MS03	Not Specified Reason Agent Generated. Reason not specified. Only to be used in case of national legislation (e.g., data protection laws) does not allow the use of AC04, RR01, RR02, RR0 and RR04. Note: limit the use of the reason code MS03 and select the appropriate reason code in the list.
RC01	Bank Identifier Incorrect. PSP identifier incorrect (i.e., invalid BIC).
RR01	Missing Debtor Account or Identification. Regulatory reason.

RR02	Missing Debtor's Name or Address. Regulatory reason. Note: This code cannot be used in certain SEPA countries for reasons of data protection. MS03 could be used as alternative.
RR03	Missing Creditor's Name or Address. Regulatory reason. Note: This code cannot be used in certain SEPA countries for reasons of data protection. MS03 could be used as alternative.
RR04	Regulatory reason. Regulatory reason. Only to be used for Regulatory Reasons other than RR01, RR02 or RR03. Note: This code cannot be used in certain SEPA countries for reasons of data protection. MS03 could be used as alternative.
TM01	Invalid Cut Off Time. Time-out-maximum execution time has been exceeded. This reason code can only be used between the Beneficiary PSP and its CSM. Reason code TM01 is not allowed in a negative confirmation message to the Originator PSP. Instead the code AB05 or AB06 can be used.

Reason Codes for Request for Cancellation (camt.055)

The **Reason** code (Undrlyg - OrgnlPmtInfAndCxl - TxInf - CxlRsnInf) explains the reason for initiating a Request for Cancellation of the request-to-pay. The values are described in the table below.

ISO CODE	NAME. DESCRIPTION
Payee reason codes	
AC02	InvalidDebtorAccountNumber. Debtor account number invalid or missing.
AM09	Wrong amount. Amount is not the amount agreed or expected.
BE16	InvalidDebtorIdentificationCode. Debtor or Ultimate Debtor identification code missing or invalid.
DT01	InvalidDate. Invalid date (for example, wrong or missing settlement date).
MODT	ModifiedTransaction. The underlying transaction in relation to an RTP was modified.
PAID	TransactionAlreadyPaid. The underlying transaction in relation to an RTP was already paid (via other means).
DUPL	Duplicate payment. Duplicate sending.
TECH	Technical problem. Cancellation requested following technical problems resulting in an erroneous transaction.
Payee's Service provider reason codes	
DRTP	DuplicationRequestToPay. Duplication of a request-to-pay message.
TECH	Technical problem. Cancellation requested following technical problems resulting in an erroneous transaction.
FRAD	Fraudulent Origin. Cancellation requested following a transaction that was originated fraudulently. The use of the FraudulentOrigin code should be governed by jurisdictions.
AM09	Wrong amount. Amount is not the amount agreed or expected.

BE16	InvalidDebtorIdentificationCode. Debtor or Ultimate Debtor identification code missing or invalid.
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3.3.2. The Reconciliation Message

Using a customizable schedule (operator users will specify a list of timestamps) the system generates a reconciliation message multiple times a day a for each Participant. In case of indirect clearing participants the reconciliation message is generated separately and sent to the direct clearing participant. The reconciliation messages will also be generated at the daily rollover event, depending on participant configuration. The reconciliation message will contain the following information:

- Settlement processing sequence of settlement transaction in IPS technical accounts
- Date and time of settlement
- Total initiated payments (number and value)
- Total payments received (number and value)
- Total payments (number and net value)
- Opening balance
- Closing balance
- List of transactions that have been settled, with the following fields:
 - Payment instruction reference (TxId)
 - End to end id (EndToEndId)
 - Date and time of the transaction processing ; set by IPS after finalizing the transaction
 - Date and time of acceptance of the transaction; parsed from AccptncDtTm tag
 - Amount of payment item and currency
 - The transaction processing sequence generated at transaction processing
 - The debtor Participant's BIC Code (DbtrAgt)
 - The creditor Participant's BIC Code (CdtrAgt)
 - The debtor's IBAN code (dbtrIBAN)
 - The creditor's IBAN code (cdtrIBAN)
 - Message type.

The identification of transactions from for inclusion in reconciliation messages is based on transaction processing timestamps. All transactions completed before the reconciliation cutoff will be included in the reconciliation message.

Monitoring

3.4.1. Monitoring Screens

The IPS solution enables a wide range of real-time monitoring screens for different components or events as well as the status of the Participants.

All monitors update their information automatically. The update period can be changed on each screen. Possible values are: one, two, three, four or five seconds.

Monitoring Participants' Connections

The Participants' connections are monitored to know their status in real time. The MMC monitors the status of the system Participants by comparing the time elapsed since the last STP function call reception with the System Participant Connection Timeout parameter.

This screen is available to both IPS Operator and the Participants. The information available in this monitoring display is presented in the table below.

	NAME	DESCRIPTION
1	Name	The name of the Participant. Displayed for the IPS Operator and for all Participants.
2	BIC	BIC code. Displayed for the IPS Operator and for all Participants.
3	AVG	Average response time. Displayed for the IPS Operator and owner Participant.
4	MAX	Maximum response time. Displayed for the IPS Operator and owner Participant.
5	Status	Participant status. Displayed for the IPS Operator and for all Participants. This can be: ONLINE – Active Participant consuming messages. OFFLINE – Active but disconnected from IPS. BLOCKED – Participant flagged as blocked. REMOVED – Participant no longer part of IPS. Participants with this status will only be shown to the IPS Operator.
6	Pending	Pending messages waiting to be consumed by participant. Displayed for the IPS Operator and owner Participant.
7	Total	Total messages consumed by participant in the current day. Displayed for the IPS Operator and owner Participant.

The average and maximum processing time of a Participant is measured as the mean or maximum value of the time interval² elapsed between two successive calls to the STP Message Receiving Function. Average and maximum processing time resets every day at Roll-over time. The

connectivity monitoring display allows to filter them according to the possible values of their statuses.

Monitoring Account Positions

The positions of the Participants' accounts are also monitored in real time. The screen is available to both the IPS Operator and the Participants, but the latter will only be able to see their own positions. Information can be filtered by currency and Participant. The available information is:

	NAME	DESCRIPTION
1	Account	Technical account identifier (BIC).
2	The current balance	Current account balance.
3	Available balance	Amount available for payments (AB).
4	Debit amount	Value of initiated and completed Inst transactions.
5	Number of debits	Number of Inst transactions initiated and completed.
6	Credit amount	Value of Inst transactions received and completed.
7	Number of credits	Number of Inst transactions received and completed.
8	Amount of processing debits	Value of Inst transactions initiated and in processing (HOLD).
9	Number of debits in processing	Number of initiated and in processing Inst transactions (HOLD).
10	Amount of processing credits	Value of Inst transactions received and in processing (HOLD).
11	Number of credits in processing	Number of receiving and in processing Inst transactions (HOLD).

Monitoring the Events of Liquidity Adjustments

Over the course of a working day, the IPS Operator can define more times in which adjustments will be performed. These events can be defined in a pattern that repeats each day of work or just for the current day. At the same time, there are two types of events: one-time events at an exact time, and window events that involve multiple executions with a certain period of time between two specified times. IPS allows to monitor these events chronologically ordered.

The screen is available to both the IPS Operator and the Participants.

The information available on the monitoring screen is:

	NAME	DESCRIPTION
1	Index	Event index. Indicates the event number of the current day.
2	Name	Event name.
3	Type	Event Type: Settlement.

	NAME	DESCRIPTION
4	Scheduled at	The time it was planned for execution. It is the time specified in defining the ad-hoc event or the time IPS calculated for execution in case of window events depending on the window frequency and the start time of the window.
5	Executed at	The actual time at which the event was executed. It is only available for events already done on that working day.
6	Status of the event	Possible values are: CREATED – Event created. SCHEDULED – Scheduled event. GENERATED MESSAGE – Event that is running. The event-specific message or messages were generated. PARTIAL COMPLETED – When event is ongoing and there are still messages sent to RTGS with no response yet. WITHOUT ACTIVITY – Executed event that did not imply sending a message. COMPLETE –for all messages IPS has received responses from RTGS and Event is executed.
7	Status of the sent message	The status of the generated messages, in the case of events that involved generating messages. For the same event (there may be more than one sent messages if liquidity adjustment is performed for multiple participants) all sent messages are listed. The state through which an interface message passes, is: Generated (REQ_GENERATED / MULTIPLE_REQ_GENERATED) – The message was generated and sent to the settlement system RTGS. Completed (COMPLETED) /Cancelled– The IPS received a response (positive/negative) from the settlement system RTGS. Processed (POSTED) – The settlement response was processed in the IPS.
8	Reference of the sent message	In the case of events that involved the generation of messages, the reference of these messages.

This monitoring screen permits the filtering of events by currency, event type, event status and request status.

Monitoring the Participant Message Statuses

The screen is available to both the IPS Operator and the Participants, but the latter will only be able to see their own messages.

The application will have a dashboard for monitoring participant messages processed during the current day, with the following information:

	NAME	DESCRIPTION
1	BIC	Participant BIC Code
2	Message Type	Breakdown of message type

3	#Count	Number of messages per breakdown
4	Accepted	Number of completed transactions with accepted status
5	Rejected	Number of rejected transactions
6	Error code/Count	Number of transactions for each error code

3.4.2. Alerts

IPS has a set of predefined alerts. Alerts are raised when certain events are captured in the system. The alerts included in the system let users, operators and Participants notify about events in the IPS. Notifications are submitted in the visual/auditory web interface or outside the application's graphical user interface, via email and SMS. The system allows the IPS operator to configure the presentation mode per alert (specifying visual/pop-up/audio/email/SMS – or all/multiple of these) with the possibility to enter user-level exceptions (modifying the 5 options).

Alerts can be assigned to a specific group (attached to a Participant), providing segregation of information. An IPS Operator user is able to view alerts generated for any group, while Participants' users are able to receive their group or generic notifications (without a specified group).

The alert definition contains an additional filter related to the type of user profile that can consume it (Administrator / Restricted). This filters limit users who can receive alert instances by comparing the selected profile type with the filter (instances will not be notified in case of mismatch).

Each alert has a statically defined level (Info / Warning / Error) that allows for better communication of severity (and implicitly the required response).

Additionally, the system provides a mapping function that allows definition of the set of users receiving each alert if the default configuration is not sufficient. This mapping feature also allows the IPS Operator to customize the presentation type for each individual user.

The attributes of an alert are:

Note: Name

Note: Level/ Severity (Info, Warning, Error)

Note: Profile type – Combo Box for selecting users that receive the alert if specific associations (mapping) are defined. Possible values are:

1. All – all users
2. Administrator – only IPS Operator's users
3. Restricted – only Participants' users

Note: Message (Description)

Note: Presentation type – allows to individually select each presentation type:

1. Audio
2. Visual
3. Popup
4. Email
5. SMS

- On/Off – Allows to enable or disable the alert. A deactivated alert is no longer evaluated and no longer generates alert instances.

Certain alerts contain additional parameters, such as upper (or lower) limits, or periodicity that set up the alert-specific evaluation mechanism. From the Alert menu, which contains the fields described in this table, the system operator can, optionally, restrict alerts to only particular profile types (ADMIN, GROUP, SUPERUSER, AUTHORIZED). The assignments are already setup by default based on alert type (for example Kafka Replication alert is by default mapped to “ADMIN” profile type since it wouldn’t make sense to present it to participants). When an alert is generated, depending on alert type, it is generated only for a specific participant (i.e. “Rejected messages” is generated only for the affected participant).

Alert Name	Profile	Alert Description
Balance above limit	All	Alert raised when the balance of an account gets above the high limit for that specific account
Balance under limit	All	Alert raised when the balance of an account gets under low limit for that specific account
Liquidity update message status	All	Liquidity update operation is pending or rejected by RTGS
Liquidity balance change	All	RTGS confirms the Participant’s IPS liquidity balance change.
Inactive Participant	All	No transactions sent by the Participant for a certain period of time • Period of time can be configured by the operator
Rejected messages	All	A number of consecutive messages were rejected in a predefined time interval for a specific Participant • The time window and number of messages can be configured by the operator
Broadcast Alert	All	Alert used to send custom message to users • This alert is manually generated by the Administrator to one or multiple Participants and it is a free text notification. Applies to Proxy Module as well.
Calendar Modified Alert	All	Alert raised when a calendar is modified, including creating new months for the next year/ or changing working to non-working days.

		• Its level is inf, no group filter. Contains information about year, month and calendar name Applies to Proxy Module as well.
Certificate Expiration Alert	All	Alert raised when a certificate is going to expire on the next days configured on the alert • The level is Warning. Number of days can be set by the operator. Applies to Proxy Module as well.
Daily Roll Over failed Alert	All	Alert raised when a daily rollover event has failed • A manual intervention is needed by the Administrators to identify the cause of the failure and for running the timetable manually. Applies to Proxy Module as well.
Event failed Alert	All	Alert raised when an event has failed • Raised on liquidity adjustment events failure. An investigation is necessary from the Administrators to determine the cause of the failure.
Participant modified alert	All	Alert raised when a participant is modified • Its level is Info, no group filter. The name of the Participant is included in the notification. By default, it is sent to all users Applies to Proxy Module as well.
Participant disconnected alert	All	A participant has gone offline.
IPS node not working alert	All	Alert raised when an IPS node is not working (for example, a funding operation can not be sent to the nodes) • Its level is Info, no group filter. The name of the Participant is included in the notification. By default, it is sent to all users
Settlement Windows Start failed Alert	All	Alert raised when a settlement window start event has failed. These are predefined moments in the timetable to identify RTGS business hours. There are two moments: settlement window start and settlement window stop. All RTGS events are

		validated to only take place inside this window of time. By default it is done manually, but we will customize it to extract this information automatically from RTGS to be online with bid requirement. Applies to Proxy Module as well.
Settlement Windows Stop failed Alert	All	Alert raised when a settlement window stop event has failed. These are predefined moments in the timetable to identify RTGS business hours. There are two moments: settlement window start and settlement window stop. All RTGS events are validated to only take place inside this window of time. Applies to Proxy Module as well.
Discrete Event Create Alert	All	Alert is raised when a single type of event is created.
Discrete Event Modify Alert	All	Alert is raised when a single type of event is modified.
Timetable modified alert	All	Alert raised when a timetable is modified Its level is Info, no group filter. The timetable name is included in the notification. By default, it is sent to all users. Applies to Proxy Module as well.
User was blocked (Status Disabled)	All	The user passed the threshold for failed login attempts Its level is Error, it is recommended to investigate the reason immediately. By default, it is sent to all users of the affected Participant, and the email alert is also sent to the email address configured at the Participant level. Applies to Proxy Module as well.
Window Event Create Alert	All	Alert is raised when a window type event is created.
Window Event Modify Alert	All	Alert is raised when a window type event is modified.
Dispute Arbitration Alert	All	Alert raised when changes regarding the arbitration process have been made

		This is raised when changes on the dispute are made during the arbitration process.
Close Dispute Alert	All	Alert raised when a dispute is closed
Dispute Claim GeneratedAlert	All	Alert raised when a claim has been made
Dispute Escalation Alert	All	Alert raised when the dispute escalates
Escalation Matrix Alert	All	Alert raised when escalation matrix is approved
Update Dispute Alert	All	Alert raised when a dispute is updated
New Dispute Alert	All	Alert raised when a new dispute is created
Dispute Overdue Alert	All	Alert raised when a dispute is marked as overdue

Alert Listing

IPS allows users to list alerts through a dedicated menu entry in MMC: Maintenance – Alerts – Generated. Operator users can see all generated alerts in the system while Participant users can see their alerts and a restricted number of general alerts (RTGS calendar updates, liquidity adjustment event schedule/execution etc.).

Generated alerts can be filtered using the following criteria:

1. Type

1. All
2. Active – alerts whose trigger condition still applies (i.e. Participant Disconnected Alert will remain active until the participant reconnects)
3. New – alerts which were not seen by the current user
4. Expired – alerts whose trigger condition no longer applies; an alert is marked as expired once the specific trigger is no longer present (for example if an alert is generated for a node going offline, then that alert remains in status “Active” until the node comes back online, and then the alert moves to status “Expired”)

2. Alerts – filter by alert name

3. Severity – All / ERROR / WARNING / INFO

4. Business Date from/to

5. Timestamp from/to

The following information is displayed for a generated alert:

1. Name

2. Severity
3. Business Date
4. Timestamp
5. Expired – Yes/No
6. Reason – description of the alert reason (i.e. “The node 10.0.100.1 went down.”)

Billing

The billing module allows the IPS Operator to define charging rates related to the system activity that will result in generating a billing report. There are three categories of rates: 1. participation rates, 2. ad hoc rates, and 3. activity rates. The billing system has a set of parameters that control the process, modifiable by the operator.

STP Interface

The IPS solution provides Participants with an STP interface to ensure interoperability between the central IPS and the Participant systems. This interface is based on the HTTPS protocol, a client-server protocol, through which Participants' applications require operations to be performed by the central IPS. In the context of the HTTPS protocol, operations are synchronous, which means that the client application initiates a request and waits for a response from the central system. Using this protocol, client applications are directly connected to the central system (even if this connection is intercepted and transferred further by a load-balancer system - HAProxy), which allows for better monitoring of the connection of the Participants to the central system.

The interface allows the Participant to initiate multiple connections (operations) in parallel to the IPS (PM) system. Since the operations provided by the STP interface are synchronous, the use of multiple parallel connections requires the use of multiple single-threaded applications running in parallel or a multi-threaded application. These options can also be combined, if necessary.

The IPS central system, through the PM, does not impose any limit on the maximum number of connections of a Participant. There is no restriction on the number of DS or SSL certificates used by a Participant that initiates parallel operations through the STP interface – can use the same certificate for all applications or may use different certificates. The only conditions are those described in the security section: Certificates must be active, unexpired, not revoked and imported into IPS.

An API for MMC functionality will be exposed for testing and automation. This API will be secured using TLS and JWT tokens.

MMC API will work with JSON messages.

3.6.1. Client Library

In order to facilitate the interfacing of client applications from the Participant with the central IPS, Montran provides a client library for the Java programming language. This library implements the communication protocol described above and can be easily integrated into client solutions. The functions provided by the STP interface dedicated to the Participants and implemented in the client library are as follows:

- Send message according to schema.
- Message reception (both SCTInst schema and additional messages such as reconciliation).
- Send confirmation message (pacs.002) for a received payment.
- Confirm message received (only for messages other than pacs.008 which is automatically confirmed using the previous function).
- Filter/List technical account positions.
- Apply the digital signature of the message to be sent to IPS.
- Verify the digital signature of messages received from IPS.

Message retransmission function is not assured in the event that the transmission of a message is not finalized successfully.

Report generation

The report generation mechanism is separated from transaction processing. Transactions are processed on the PM component.

Report requests are initialized from MMC and sent to a queue to a separate module, the Reporting application. The Reporting module is a standalone java application. This module generates the reports by querying the main database and sends back the response to MMC for storing, displaying, and downloading.

The main database is not used by PM for transaction processing therefore it will not impact system performance. Transactions are written to the main database through an asynchronous process, which might be delayed if there is heavy usage of the reporting functionality, but it will not impact performance. For this reason, there are no parameters to turn on or off the generation of on demand reports.

Customized reports will be managed via third party tool, i.e., Microsoft SQL Server Reporting Services.